

EXERCISES

LEVEL I

(Problems Based on Fundamentals)

ABC OF INVERSE FUNCTION

1. A function $f: R \rightarrow R$ is defined as $f(x) = 3x + 5$. Find $f^{-1}(x)$.
2. A function $f: (0, \infty) \rightarrow (2, \infty)$ is defined as $f(x) = x^2 + 2$. Then find $f^{-1}(x)$.
3. A function $f: R^+ \rightarrow [0, 1)$ is defined as $f(x) = \frac{x^2}{x^2 + 1}$. Then find $f^{-1}(x)$.
4. A function $f: [1, \infty) \rightarrow [1, \infty)$ is defined as $f(x) = 2^{x(x-1)}$. Find $f^{-1}(x)$.
5. If a function f is bijective such that $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$, then find $f^{-1}(x)$.
6. A function $f: R \rightarrow R$ is defined as $f(x) = x + \sin x$. Find $f^{-1}(x)$.
7. A function $f: [2, \infty) \rightarrow [5, \infty)$ is defined as $f(x) = x^2 - 4x + 9$. Find its inverse.
8. Find all the real solutions to the equation

$$x^2 - \frac{1}{4} = \sqrt{x + \frac{1}{4}}$$

9. A function f is defined as $f(x) = 3x + 5$ where $f: R \rightarrow R$, then find $f^{-1}(x)$.
10. A function f is defined as $f(x) = \frac{x}{x-1}$ where $f: R - \{1\} \rightarrow R - \{1\}$, then find $f^{-1}(x)$.
11. A function f is defined as $f(x) = \frac{1}{x^2 + 1}$ where $f: R^+ \cup \{0\} \rightarrow (0, 1]$, find $f^{-1}(x)$.
12. A function f is bijective such that $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$, then find $f^{-1}(x)$.

13. A function $f: [1, 1] \rightarrow \left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right]$ is defined as $f(x) = \frac{x}{x^2 + 1}$, then find $f^{-1}(x)$.

ABC OF INVERSE TRIGONOMETRIC FUNCTIONS

14. Find the domain of $f(x) = \sin^{-1}(3x + 5)$.
15. Find the domain of $f(x) = \sin^{-1}\left(\frac{x}{x+1}\right)$.
16. Find the domain of $f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$.
17. Find the domain of $f(x) = \sin^{-1}\left(\frac{|x|}{2}\right)$.
18. Find the domain of $f(x) = \sin^{-1}(\log_2 x)$.
19. Find the domain of $f(x) = \sin^{-1}(\log_4 x^2)$.
20. Solve for x and y : $\sin^{-1} x + \sin^{-1} y = \pi$.
21. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$,

then find the value of

$$\frac{x^{2013} + y^{2013} + z^{2013}}{x^{2014} + y^{2014} + z^{2014}} \cdot 9$$

22. Find the range of $f(x) = 2 \sin^{-1}(3x + 5) + \frac{\pi}{4}$.
23. Solve the inequality: $\sin^{-1} x > \sin^{-1}(3x - 1)$.
24. Find the domain of $f(x) = \cos^{-1}(2x + 4)$.
25. Find the range of $f(x) = 2 \cos^{-1}(3x + 5) + \frac{\pi}{4}$.
26. Find the range of $f(x) = 3 \cos^{-1}(x^2) - \frac{\pi}{2}$.
27. Solve for x : $\cos^{-1} x + \cos^{-1} x^2 = 0$.
28. Solve for x : $[\sin^{-1} x] + [\cos^{-1} x] = 0$, where x is a non-negative real number and $[.]$ denotes the greatest integer function.
29. Find the domain of $f(x) = \cos^{-1}\left(\frac{x^2}{x^2 + 1}\right)$.
30. Solve for x : $\cos^{-1}(x) > \cos^{-1}(x^2)$.
31. Find the domain of $f(x) = \tan^{-1}(\sqrt{9 - x^2})$.
32. Find the range of the function

$$f(x) = 2 \tan^{-1}(1 - x^2) + \frac{\pi}{6}$$

33. Find the range of $f(x) = \cot^{-1}(2x - x^2)$.
34. Solve for x : $[\cot^{-1} x] + [\cos^{-1} x] = 0$.
35. Find the number of solutions of $\sin\{x\} = \cos\{x\}$, $\forall x \in [0, 2\pi]$.

Q. Find the domains of each of the following functions:

36. $f(x) = \sin^{-1}\left(\frac{|x|}{3}\right) + \cos^{-1}\left(\frac{1 - |x|}{4}\right)$
37. $f(x) = \sin^{-1}(2x^2 - 1)$
38. $f(x) = \sqrt{5\pi \sin^{-1} x - 6(\sin^{-1} x)^2}$
39. $f(x) = \log_2\left(\frac{3 \tan^{-1} x + \pi}{\pi - 4 \tan^{-1} x}\right)$
40. $f(x) = \cos^{-1}\left(\frac{3}{2 + \sin x}\right)$
41. $f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$
42. $f(x) = \cos^{-1}\left(\frac{x^2 + 1}{x^2}\right)$
43. $f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$
44. $f(x) = \sin^{-1}\left(\log_2\left(\frac{x^2}{2}\right)\right)$
45. $f(x) = \sin^{-1}[2 - 3x^2]$
46. $f(x) = \frac{1}{x} + 3 \sin^{-1} x + \frac{1}{\sqrt{x-2}}$
47. $f(x) = \sin^{-1}(\log_2 x^2)$
48. $f(x) = e^x + \sin^{-1}\left(\frac{x}{2} - 1\right) + \frac{1}{x}$

$$49. f(x) = \sqrt{\sin^{-1}(\log_x 2)}$$

$$50. f(x) = \sqrt{\sin^{-1}(\log_2 x)}$$

Q. Find the ranges of each of the following functions:

$$51. f(x) = \sin^{-1}(2x - 3)$$

$$52. f(x) = 2 \sin^{-1}(2x - 1) - \frac{\pi}{4}$$

$$53. f(x) = 2 \cos^{-1}(-x^2) - \pi$$

$$54. f(x) = \frac{1}{2} \tan^{-1}(1 - x^2) - \frac{\pi}{4}$$

$$55. f(x) = \cot^{-1}(2x - x^2)$$

$$56. f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$$

$$57. f(x) = \sin^{-1} x + \sec^{-1} x + \tan^{-1} x$$

$$58. f(x) = 3 \cot^{-1} x + 2 \tan^{-1} x + \frac{\pi}{4}$$

$$59. f(x) = \operatorname{cosec}^{-1}[1 + \sin^2 x]$$

$$60. f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$$

CONSTANT PROPERTY

$$61. \text{ Find the range of } f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$$

$$62. \text{ Solve for } x: 4 \sin^{-1}(x - 2) + \cos^{-1}(x - 2) = \pi$$

$$63. \text{ Solve for } x: \sin^{-1}(x^2 - 2x + 1) + \cos^{-1}(x^2 - x) = \frac{\pi}{2}$$

$$64. \text{ Find the number of real solutions of } \tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

$$65. \text{ If } \sin^{-1} \left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \right) + \cos^{-1} \left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \right) = \frac{\pi}{2}, \text{ for } 0 < |x| < \sqrt{2}, \text{ then find } x.$$

$$66. \text{ Solve for } x: \sin^{-1} x > \cos^{-1} x$$

Q. Solve for x:

$$67. (\sin^{-1} x)^2 - 3 \sin^{-1} x + 2 = 0$$

$$68. \sin^{-1} x + \sin^{-1} 2y = \pi$$

$$69. \cos^{-1} x + \cos^{-1} x^2 = 2\pi$$

$$70. \cos^{-1} x + \cos^{-1} x^2 = 0$$

$$71. 4 \sin^{-1}(x - 1) + \cos^{-1}(x - 1) = \pi$$

$$72. \cot^{-1} \left(\frac{1}{x^2 - 1} \right) + \tan^{-1}(x^2 - 1) = \frac{\pi}{2}$$

$$73. \cot^{-1} \left(\frac{x^2 - 1}{2x} \right) + \tan^{-1} \left(\frac{2x}{x^2 - 1} \right) = \frac{2\pi}{3}$$

$$74. 4 \sin^{-1} x + \cos^{-1} x = \frac{3\pi}{4}$$

$$75. 5 \tan^{-1} x + 3 \cot^{-1} x = \frac{7\pi}{4}$$

$$76. 5 \tan^{-1} x + 4 \cot^{-1} x = 2\pi$$

$$77. \cot^{-1} x - \cot^{-1}(x+1) = \frac{\pi}{2}$$

$$78. [\sin^{-1} x] + [\cos^{-1} x] = 0$$

$$79. [\tan^{-1} x] + [\cot^{-1} x] = 0$$

$$80. [\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 0$$

$$81. [\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 1$$

$$82. (\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$$

CONVERSION OF INVERSE TRIGONOMETRIC FUNCTIONS

$$83. \text{ Find the value of } \cos \left(\frac{1}{2} \cos^{-1} \left(\frac{3}{5} \right) \right).$$

$$84. \text{ Find the value of } \sin \left(\frac{\pi}{4} + \sin^{-1} \left(\frac{1}{2} \right) \right)$$

$$85. \text{ If } m \text{ is a root of } x^2 + 3x + 1 = 0, \text{ then find the value of } \tan^{-1}(m) + \tan^{-1} \left(\frac{1}{m} \right).$$

$$86. \text{ Prove that } \cos(\tan^{-1}(\sin(\cot^{-1} x))) = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

Q. Solve for x:

$$87. 6(\sin^{-1} x)^2 - \pi \sin^{-1} x \leq 0$$

$$88. \frac{2 \tan^{-1} x + \pi}{4 \tan^{-1} x - \pi} \leq 0$$

$$89. \sin^{-1} x < \sin^{-1} x^2$$

$$90. \cos^{-1} x > \cos^{-1} x^2$$

$$91. \log^2(\tan^{-1} x) > 1$$

$$92. (\cot^{-1} x)^2 - 5 \cot^{-1} x + 6 > 0$$

$$93. \sin^{-1} x < \cos^{-1} x$$

$$94. \sin^{-1} x > \sin^{-1}(1 - x)$$

$$95. \sin^{-1} 2x > \operatorname{cosec}^{-1} x$$

$$96. \tan^{-1} 3x < \cot^{-1} x$$

$$97. \cos^{-1} 2x^3 \sin^{-1} x$$

$$98. x^2 - 2x < \sin^{-1}(\sin 2)$$

$$99. \sin^{-1} \left(\frac{x}{2} \right) < \cos^{-1}(x+1)$$

$$100. \tan^{-1} 2x > 2 \tan^{-1} x$$

$$101. \tan(\cos^{-1} x) \leq \sin \left(\cot^{-1} \left(\frac{1}{2} \right) \right)$$

COMPOSITION OF TRIGONOMETRIC FUNCTIONS AND ITS INVERSE

102. Let $f(x) = \sin^{-1} x + \cos^{-1} x$
Then find the value of:

$$(i) f \left(\frac{1}{m^2 + 1} \right), m \in R$$

$$(ii) f \left(\frac{m^2}{m^2 + 1} \right), m \in R$$

$$(iii) f \left(\frac{m}{m^2 + 1} \right), m \in R$$

$$(iv) f(m^2 - 2m + 6), m \in R$$

$$(v) f(m^2 + 1), m \in R$$

103. If $\cos^{-1}x + \cos^{-1}y = \frac{2\pi}{3}$, then find the value of $\sin^{-1}x + \sin^{-1}y$.

104. If m is the root of $x^2 + 3x + 1 = 0$, then find the value of $\tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right)$.

105. Solve for x :

$$\sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+2}\right)\right) > \sin^{-1}(\sin 3)$$

COMPOSITION OF INVERSE TRIGONOMETRIC FUNCTIONS AND TRIGONOMETRIC FUNCTIONS

106. Find the values of:

- (i) $\sin^{-1}(\sin 3)$ (ii) $\sin^{-1}(\sin 53)$
(iii) $\sin^{-1}(\sin 7)$ (iv) $\sin^{-1}(\sin 10)$
(v) $\sin^{-1}(\sin 20)$

107. Find the values of:

- (i) $\cos^{-1}(\cos 2)$ (ii) $\cos^{-1}(\cos 3)$
(iii) $\cos^{-1}(\cos 5)$ (iv) $\cos^{-1}(\cos 7)$
(v) $\cos^{-1}(\cos 10)$

108. Find the values of:

- (i) $\tan^{-1}(\tan 3)$ (ii) $\tan^{-1}(\tan 5)$
(iii) $\tan^{-1}(\tan 7)$ (iv) $\tan^{-1}(\tan 10)$
(v) $\tan^{-1}(\tan 15)$

109. Find the value of $\cos^{-1}(\sin(-5))$

110. Find $f'(x)$, where $f(x) = \sin^{-1}(\sin x)$ and $-2\pi \leq x \leq \pi$

111. Find $f'(x)$, where $f(x) = \cos^{-1}(\cos x)$ and $-\pi \leq x \leq 2\pi$

112. Solve for x : $\sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+1}\right)\right) < \pi - 3$

113. Find the integral values of x satisfying the inequality, $x^2 - 3x < \sin^{-1}(\sin 2)$

114. Find the value of

$$\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) + \tan^{-1}(\tan 50)$$

Q. Find the values of:

115. $\sin^{-1}(\sin 1) + \sin^{-1}(\sin 2) + \sin^{-1}(\sin 3)$

116. $\sin^{-1}(\sin 10) + \sin^{-1}(\sin 20)$
 $+ \sin^{-1}(\sin 30) + \sin^{-1}(\sin 40)$

117. $\cos^{-1}(\cos 1) + \cos^{-1}(\cos 2)$
 $+ \cos^{-1}(\cos 3) + \cos^{-1}(\cos 4)$

118. $\cos^{-1}(\cos 10) + \cos^{-1}(\cos 20)$
 $+ \cos^{-1}(\cos 30) + \cos^{-1}(\cos 40)$

119. $\sin^{-1}(\sin 10) + \cos^{-1}(\cos 10)$

120. $\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50)$

121. $\sin^{-1}(\sin 100) + \cos^{-1}(\cos 100)$

122. $\cos^{-1}(\sin(-5)) + \sin^{-1}(\cos(-5))$

123. Find the number of ordered pairs of (x, y) satisfying the equations $y = |\sin x|$ and $y = \cos^{-1}(\cos x)$, where $x \in [-2\pi, 2\pi]$

124. Let $f(x) = \cos^{-1}(\cos x) - \sin^{-1}(\sin x)$ in $[0, \pi]$. Find the area bounded by $f(x)$ and x -axis.

125. $\tan^{-1}(\tan 1) + \tan^{-1}(\tan 2)$
 $+ \tan^{-1}(\tan 3) + \tan^{-1}(\tan 4)$

126. $\tan^{-1}(\tan 20) + \tan^{-1}(\tan 40)$
 $+ \tan^{-1}(\tan 60) + \tan^{-1}(\tan 80)$

127. $\sin^{-1}(\sin 15) + \cos^{-1}(\cos 15) + \tan^{-1}(\tan 15)$

128. $\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) - \tan^{-1}(\tan 50)$

129. $3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$

130. $\sin^{-1}\left(\sin\left(\frac{2x^2+4}{x^2+1}\right)\right) < \pi - 3$

SUM OF ANGLES

131. Find the value of $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$.

132. Find the value of

$$\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)$$

133. Find the value of $\tan^{-1}(9) + \tan^{-1}\left(\frac{5}{4}\right)$.

134. Find the value of

$$\sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) - \sin^{-1}\left(\frac{63}{65}\right)$$

135. Prove that $2 \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$

136. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, prove that

$$x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$$

137. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$
prove that $x^2 + y^2 + z^2 + 2xyz = 1$

138. If $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$, prove that

$$9x^2 + 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta$$

139. Let $m = \frac{\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)}{\cot^{-1}(1) + \cot^{-1}(2) + \cot^{-1}(3)}$

Then find the value of $(m-1)^{2013}$.

140. Solve for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{3\pi}{4}$

141. Solve for x : $\sin^{-1}(x) + \sin^{-1}(2x) = \frac{\pi}{3}$

142. Let $f(x) = \cos^{-1}(x)$

$$+ \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right), \text{ for } \frac{1}{2} \leq x \leq 1$$

Then find $f(2013)$

143. $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$

144. $x^2 - 4x > \sin^{-1}(\sin(\pi^{3/2})) + \cos^{-1}(\cos[\pi^{3/2}])$

145. $\cos(\tan^{-1}x) = x$

146. $\sin(\tan^{-1}x) = \cos(\cot^{-1}(x+1))$

147. $\sec^{-1}\left(\frac{x}{2}\right) - \sec^{-1}x = \sec^{-1}2$

148. $\cos\left(\tan^{-1}\left(\cot\left(\sin^{-1}\left(x + \frac{3}{2}\right)\right)\right)\right) + \tan(\sec^{-1}x) = 0$

149. Find the smallest +ve integer x so that

$$\tan\left(\tan^{-1}\left(\frac{x}{10}\right) + \tan^{-1}\left(\frac{1}{x+1}\right)\right) = \tan\left(\frac{\pi}{4}\right)$$

150. Find the least integral value of k for which $(k-2)x^2 + 8x + k + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$ holds for all x in R .

151. If $\alpha = 2 \tan^{-1}\left(\frac{1+x}{1-x}\right)$ and $\beta = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for $0 < x < 1$, then prove that $\alpha + \beta = \pi$.

152. Let $f(x) = \sin^{-1}(\sin x)$, $\forall x \in [-\pi, 2\pi]$. Then find $f'(x)$.

153. Let $f(x) = \cos^{-1}(\cos x)$, $\forall x \in [-2\pi, \pi]$. Then find $f'(x)$.

154. Let, $f(x) = \tan^{-1}(\tan x)$, $\forall x \in \left[-\frac{3\pi}{2}, \frac{5\pi}{2}\right]$. Then find $f'(x)$.

155. Prove that $\sin^{-1}\left(\frac{1}{5}\right) + \cot^{-1}(3) = \frac{\pi}{4}$.

156. Prove that $2 \tan^{-1}\left(\frac{3}{2}\right) + \tan^{-1}\left(\frac{12}{5}\right) = \pi$.

Q. Find the simplest form of:

157. $\cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right)$

158. $\sin^{-1}(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2})$

159. $\sin^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right)$, $-\frac{\pi}{4} < x < \frac{\pi}{4}$

160. $\cos^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right)$, $\frac{\pi}{4} < x < \frac{5\pi}{4}$

161. $\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right)$

162. $\sin^{-1}\left(\frac{3}{5}\cos x + \frac{4}{5}\sin x\right)$

MULTIPLE ANGLES

163. Find the value of $\sin\left(2 \sin^{-1}\left(\frac{1}{4}\right)\right)$

164. Find the value of $\cos\left(2 \cos^{-1}\left(\frac{1}{3}\right)\right)$

165. Find the value of $\cos\left(2 \tan^{-1}\left(\frac{1}{3}\right)\right)$

166. Find the value of $\sin\left(\frac{1}{2} \cot^{-1}\left(\frac{3}{4}\right)\right)$

167. Find the value of $\tan^{-1}\left(\frac{3\pi}{4} - 2 \tan^{-1}\left(\frac{3}{4}\right)\right)$

168. Prove that $\sin\left(2 \sin^{-1}\left(\frac{1}{2}\right)\right) = \frac{\sqrt{3}}{2}$

169. Prove that $\sin\left(3 \sin^{-1}\left(\frac{1}{3}\right)\right) = \frac{23}{27}$

170. Prove that $\cos\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{8}\right)\right) = \frac{3}{4}$

171. Prove that $\cos\left(\frac{1}{2} \cos^{-1}\left(-\frac{1}{10}\right)\right) = \frac{3\sqrt{5}}{10}$

172. Prove that $\sin\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{9}\right)\right) = \frac{2}{3}$

173. Prove that $\sin\left(\frac{1}{4} \tan^{-1}(\sqrt{63})\right) = \frac{1}{2\sqrt{2}}$

174. Prove that $\cos\left(\frac{1}{4} \left(\tan^{-1}\left(\frac{24}{7}\right)\right)\right) = \frac{3}{\sqrt{10}}$

175. Prove that $\tan\left(\frac{1}{2} \cos^{-1}\left(\frac{2}{3}\right)\right) = \frac{1}{\sqrt{5}}$

176. Prove that $\tan\left(2 \tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right) = -\frac{7}{17}$

177. $\tan\left(\frac{3\pi}{4} - \frac{1}{4} \sin^{-1}\left(-\frac{4}{5}\right)\right) = \frac{1-\sqrt{5}}{2}$

178. Find the integral values of x satisfying the inequation $x^2 - 3x < \sin^{-1}(\sin 2)$

179. Find the value of x satisfying the inequation $3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$

180. For what value of x ,

$$f(x) = \cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{1-3x^2}}{2}\right\}$$

is a constant function.

MORE MULTIPLE ANGLES

181. Let $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + 2 \tan^{-1}(x)$, $x > 1$

Then find the value of $f(2013)$

182. Let $f(x) = 2 \tan^{-1}\left(\frac{1+x}{1-x}\right) + \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

for $0 \leq x < 1$. Then find the value of $f\left(\frac{1}{2014}\right)$

183. Let $f(x) = \sin^{-1}\left(\frac{6x}{x^2+9}\right) + 2 \tan^{-1}\left(-\frac{x}{3}\right)$

is independent of x , then find the value of x

184. Find the interval of x for which the function

$f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + 2 \tan^{-1}(x)$ is a constant function.

185. Find the interval of x for which the function $f(x) = 3 \cos^{-1}(2x^2 - 1) + 2 \cos^{-1}(4x^3 - 3x)$ is independent of x

186. If $\tan^{-1}y : \tan^{-1}x = 4 : 1$, then express y as algebraic function of x . Also, prove that $\tan\left(22\frac{1}{2}^\circ\right)$ is a root of $x^4 - 6x^2 + 1 = 0$

BOARD SPECIAL PROBLEMS

Q. Prove that:

187. $\tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2 \right)$
188. $\tan \left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} \frac{a}{b} \right) + \tan \left(\frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{a}{b} \right) = \frac{2b}{a}$
189. $\tan^{-1} \left(\frac{p-q}{1+pq} \right) + \tan^{-1} \left(\frac{q-r}{1+qr} \right) + \tan^{-1} \left(\frac{r-p}{1+pr} \right) = \pi$
where $p > q > 0$ and $pr < 1 < qr$
190. $\cot^{-1} \left(\frac{ab+1}{a-b} \right) + \cot^{-1} \left(\frac{bc+1}{b-c} \right) + \cot^{-1} \left(\frac{ca+1}{c-a} \right) = 0$
191. $\tan \left(\frac{1}{2} \sin^{-1} \left(\frac{2x}{1+x^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right) = \left(\frac{x+y}{1-xy} \right), xy < 1$
192. $\tan^{-1} \left(\frac{1-x}{1+x} \right) - \tan^{-1} \left(\frac{1-y}{1+y} \right) = \sin^{-1} \left(\frac{y-x}{\sqrt{(1+x^2)(1+y^2)}} \right)$
193. $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = 0$
194. $2 \tan^{-1} \left(\sqrt{\frac{a-b}{a+b}} \tan \left(\frac{\theta}{2} \right) \right) = \cos^{-1} \left(\frac{b+a \cos \theta}{a+b \cos \theta} \right)$
195. $\tan (2 \tan^{-1} a) = 2 \tan (\tan^{-1} a + \tan^{-1} a^3)$
196. $\cos^{-1} x + \cos^{-1} \left(\frac{x + \sqrt{3-3x^2}}{2} \right) = \frac{\pi}{3}, \frac{1}{2} < x < 1$
197. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$, then prove that
 $x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$
198. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$, then prove that
 $x^2 + y^2 + z^2 + 2xyz = 1$
199. If $\cos^{-1} \left(\frac{x}{2} \right) + \cos^{-1} \left(\frac{y}{3} \right) = \theta$, then prove that
 $9x^2 - 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta$.
200. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then prove that
 $x^2 + y^2 + z^2 - 2xyz = 1$.
201. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then prove that
 $xy + yz + zx = 3$.
202. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then find the value of
 $\frac{x^{2012} + y^{2012} + z^{2012}}{x^{2013} + y^{2013} + z^{2013}} = \frac{9}{9}$
203. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then prove that, $xy + yz + zx = 3$.

204. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of

$$\left(\frac{x^{2013} + y^{2013} + z^{2013} + 6}{x^{2014} + y^{2014} + z^{2014}} \right)$$

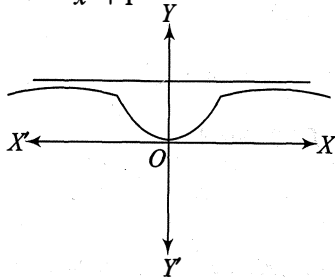
205. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$, then prove that $xy + yz + zx = 1$.
206. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$, then prove that $x + y + xy = 1$.
207. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, then prove that $x + y + z = xyz$.
208. If $\tan^{-1} \left(\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right) = \alpha$, then prove that
 $x^2 = \sin 2\alpha$
209. Let $m = \tan^2 (\sec^{-1} 2) + \cot^2 (\operatorname{cosec}^{-1} 3)$. Then find the value of $(m^2 + m + 10)$.
210. If $\frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = \frac{\pi}{4}$, then find the value of $\tan \theta$.
211. Let $m = \frac{(\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3)}{(\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3)}$, then prove that
 $(m+2)^{m+1} = 64$.

Q. Solve for x:

212. $\tan^{-1} (2x) + \tan^{-1} (3x) = \frac{3\pi}{4}$
213. $\tan^{-1} \left(\frac{x+1}{x-1} \right) + \tan^{-1} \left(\frac{x-1}{x} \right) = \tan^{-1} (-7)$
214. $\sin^{-1} (2x) + \sin^{-1} (x) = \frac{\pi}{3}$
215. $\sin^{-1} \left(\frac{1}{\sqrt{5}} \right) + \cos^{-1} x = \frac{\pi}{4}$
216. $\sin^{-1} (x) + \sin^{-1} (3x) = \frac{\pi}{3}$
217. $\tan^{-1} \left(\frac{1}{1+2x} \right) + \tan^{-1} \left(\frac{1}{1+4x} \right) = \tan^{-1} \left(\frac{2}{x^2} \right)$
218. $2 \tan^{-1} (2x+1) = \cos^{-1} x$
219. $\cos^{-1} x - \sin^{-1} x = \cos^{-1} (x\sqrt{3})$
220. If $\tan^{-1} y : \tan^{-1} x = 4 : 1$, express y as an algebraic function of x . Hence, prove that $\tan \left(\frac{\pi}{8} \right)$ is a root of
 $x^4 + 1 = 6x^2$

Hints & Solutions

3. Given, $f(x) = \frac{x^2}{x^2+1}$



$$\Rightarrow f(x) = 1 - \frac{1}{x^2+1}$$

$$\Rightarrow f'(x) = \frac{2x}{(x^2+1)^2} > 0, \forall x \in \mathbb{R}^+$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, let $y = \frac{x^2}{x^2+1}$

$$\Rightarrow y \cdot x^2 + y = x^2$$

$$\Rightarrow x^2(y-1) = -y$$

$$\Rightarrow x^2 = -\frac{y}{(y-1)} = \frac{y}{(1-y)}$$

$$\Rightarrow x = \sqrt{\frac{y}{(1-y)}}$$

$$\Rightarrow R_f = (0, 1) = \text{Co-domain}$$

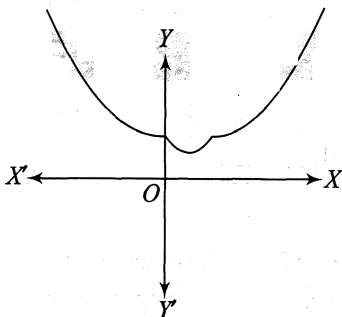
$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

$\Rightarrow f^{-1}(x)$ is exists.

$$\text{Hence, } f^{-1}(x) = \sqrt{\frac{x}{1-x}}$$

4. Given, $f(x) = 2^{x(x-1)}$



$$\Rightarrow f'(x) = 2^{x(x-1)} \times (2x-1) \times \log 2 > 0$$

for all x in $[1, \infty)$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, $R_f = [1, \infty)$

$$\Rightarrow R_f = [1, \infty) = \text{Co-domain}$$

$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

So its inverse is exists.

Let $y = 2^{x^2-x}$

$$\Rightarrow y = 2^{x^2-x}$$

$$\Rightarrow x^2 - x = \log_2(y)$$

$$\Rightarrow x^2 - x - \log_2(y) = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1+4\log_2(y)}}{2}$$

$$\Rightarrow x = \frac{1 + \sqrt{1+4\log_2(y)}}{2}$$

$$\text{Thus, } f^{-1}(x) = \frac{1 + \sqrt{1+4\log_2(x)}}{2}$$

5. Since f is a bijective function, so its inverse exists.

Let $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}} = \frac{10^{2x} - 1}{10^{2x} + 1}$

$$\Rightarrow y \times 10^{2x} + y = 10^{2x} - 1$$

$$\Rightarrow 10^{2x}(y-1) = -y-1$$

$$\Rightarrow 10^{2x} = \frac{y+1}{y-1} = \frac{y+1}{1-y}$$

$$\Rightarrow 2x = \log_{10} \left(\frac{y+1}{1-y} \right)$$

$$\Rightarrow x = \frac{1}{2} \log_{10} \left(\frac{1+y}{1-y} \right)$$

$$\text{Thus, } f^{-1}(x) = \frac{1}{2} \log_{10} \left(\frac{1+x}{1-x} \right)$$

6. Given, $f(x) = x + \sin x$

$$\Rightarrow f'(x) = 1 + \cos x \geq 0 \text{ for all } x \text{ in } \mathbb{R}.$$

$\Rightarrow f$ is strictly increasing function

$\Rightarrow f$ is a one one function.

Also, the range of a function is $\mathbb{R}$.

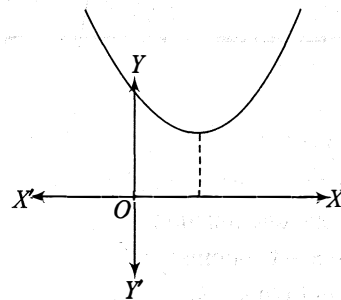
$\Rightarrow f$ is a onto function

Thus, f is a bijective function.

Hence, f^{-1} exists.

Therefore, $f^{-1}(x) = x - \sin x$

7. Given, $f(x) = x^2 - 4x + 9$



$$\Rightarrow f'(x) = 2x - 4 \geq 0 \text{ for all } x \text{ in } D_f$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, $R_f = [5, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Therefore, f is a bijective function.

Hence, its inverse exists.

$$\text{Let } y = x^2 - 4x + 3$$

$$\Rightarrow x^2 - 4x + (5 - y) = 9$$

$$\Rightarrow x = \frac{4 \pm \sqrt{16 - 4(5 - y)}}{2}$$

$$\Rightarrow x = \frac{4 \pm \sqrt{4y + 16 - 20}}{2}$$

$$\Rightarrow x = \frac{4 \pm \sqrt{4(y - 1)}}{2} = 2 \pm \sqrt{(y - 1)}$$

$$\Rightarrow x = 2 + \sqrt{(y - 1)}, \text{ since } x \geq 2$$

$$\Rightarrow f^{-1}(x) = 2 + \sqrt{(x - 1)}$$

8. Consider the function

$$f: [0, \infty) \rightarrow \left[-\frac{1}{4}, \infty\right), \text{ where}$$

$$f(x) = x^2 - \frac{1}{4}$$

Clearly, f is a one one and onto function.

So its inverse exists.

$$\text{Let its inverse be } f^{-1}: \left[-\frac{1}{4}, \infty\right) \rightarrow [0, \infty).$$

$$\Rightarrow f^{-1}(x) = \sqrt{x + \frac{1}{4}}.$$

Consequently, we can say that, the two sides of the given equation are inverse to each other.

Thus, the intersection point is the solution of the given equation. $f(x) = x$

$$\Rightarrow x^2 - \frac{1}{4} = x$$

$$\Rightarrow x^2 - x = \frac{1}{4}$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\Rightarrow \left(x - \frac{1}{2}\right) = \pm \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{2} \pm \frac{1}{2}$$

Hence, the solutions are

$$\left\{\frac{1}{2} + \frac{1}{2}, \frac{1}{2} - \frac{1}{2}\right\}$$

9. Clearly, f is bijective

So, its inverse exists

$$\text{Let } y = 3x + 5$$

$$\Rightarrow x = \frac{y - 5}{3}$$

$$\text{Thus, } f^{-1}(x) = \frac{x - 5}{3}$$

10. Clearly, f is bijective

So, its inverse exists

$$\text{Let } y = \frac{x}{x - 1}$$

$$\Rightarrow xy - y = x$$

$$\Rightarrow x(y - 1) = y$$

$$\Rightarrow x = \frac{y}{(y - 1)}$$

$$\text{Thus, } f^{-1}(x) = \frac{x}{(x - 1)}$$

11. Clearly, f is bijective

So its inverse exists

$$\text{Let } y = x^2 + 1$$

$$\Rightarrow x = \sqrt{y - 1}$$

$$\text{Thus, } f^{-1}(x) = \sqrt{x - 1}$$

12. Since f is bijective, so its inverse exists

$$\text{Let } y = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$$

$$\Rightarrow y = \frac{2^x - 2^{-x}}{1 + 2^x + 2^{-x}}$$

$$\Rightarrow \frac{y + 1}{y - 1} = \frac{2^x - 2^{-x} + 2^x + 2^{-x}}{2^x - 2^{-x} - 2^x - 2^{-x}}$$

$$\Rightarrow \frac{y + 1}{y - 1} = -\frac{2 \cdot 2^x}{2 \cdot 2^{-x}}$$

$$\Rightarrow 2^{2x} = \frac{y + 1}{1 - y}$$

$$\Rightarrow 2x = \log_2 \left(\frac{y + 1}{1 - y} \right)$$

$$\Rightarrow x = \frac{1}{2} \log_2 \left(\frac{y + 1}{1 - y} \right)$$

$$\text{Thus, } f^{-1}(x) = \frac{1}{2} \log_2 \left(\frac{x + 1}{1 - x} \right)$$

13. Clearly, f is bijective.

So, its inverse exists

$$\text{Let } y = \frac{x}{x^2 + 1}$$

$$\Rightarrow yx^2 - x + y = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 - 4y^2}}{2y}$$

$$\text{Thus, } f^{-1}(x) = \frac{1 + \sqrt{1 - 4x^2}}{2x}$$

14. We have

$$1 \leq 3x + 5 \leq 1$$

$$\Rightarrow 6 \leq 3x \leq 1$$

$$\Rightarrow -2 \leq x \leq -\frac{4}{3}$$

$$\Rightarrow D_f = x \in \left[-2, -\frac{4}{3}\right]$$

15. We have

$$-1 \leq \frac{x}{x+1} \leq 1$$

Case I: When $\frac{x}{x+1} \leq 1$

$$\Rightarrow \frac{x}{x+1} - 1 \leq 0$$

$$\Rightarrow \frac{-1}{x+1} \leq 0$$

$$\Rightarrow \frac{1}{x+1} \geq 0$$

$$\Rightarrow x > -1$$

Case II: When $\frac{x}{x+1} \geq -1$

$$\Rightarrow \frac{x}{x+1} + 1 \geq 0$$

$$\Rightarrow \frac{2x+1}{x+1} \geq 0$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[-\frac{1}{2}, \infty\right)$$

$$\text{Hence, } D_f = \left[-\frac{1}{2}, \infty\right).$$

16. We have

$$-1 \leq \frac{x^2+1}{2x} \leq 1$$

$$\Rightarrow \left| \frac{x^2+1}{2x} \right| \leq 1$$

$$\Rightarrow \frac{|x^2+1|}{|2x|} \leq 1$$

$$\Rightarrow \frac{|x^2+1|}{2|x|} \leq 1$$

$$\Rightarrow x^2+1 \leq 2|x|$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow (|x| - 1)^2 \leq 0$$

$$\Rightarrow (|x| - 1)^2 = 0$$

$$\Rightarrow (|x| - 1) = 0$$

$$\Rightarrow |x| = 1$$

$$\Rightarrow x = \pm 1$$

$$\text{Hence, } D_f = \{-1, 1\}$$

17. We have $-1 \leq \frac{|x|-1}{2} \leq 1$

$$\Rightarrow -2 \leq |x| - 1 \leq 2$$

$$\Rightarrow -1 \leq |x| \leq 3$$

$$\Rightarrow |x| \leq 3 \quad (\because |x|^3 - 1 \text{ is rejected})$$

$$\Rightarrow -3 \leq x \leq 3$$

$$\text{Hence, } D_f = [-3, 3]$$

18. We have $-1 \leq (\log_2 x) \leq 1$

$$\Rightarrow 2^{-1} \leq x \leq 2^1$$

$$\Rightarrow \frac{1}{2} \leq x \leq 2$$

$$\text{Hence, } D_f = \left[\frac{1}{2}, 2\right]$$

19. We have

$$-1 \leq \log_3 x^2 \leq 1$$

$$\Rightarrow 4^{-1} \leq x^2 \leq 4^1$$

$$\Rightarrow \frac{1}{4} \leq x^2 \leq 4$$

$$\Rightarrow \frac{1}{2} \leq |x| \leq 2$$

$$\Rightarrow |x| \leq 2 \text{ and } |x| \geq \frac{1}{2}$$

$$\Rightarrow -2 \leq x \leq 2 \text{ and } x \geq \frac{1}{2} \text{ and } x \leq -\frac{1}{2}$$

$$\Rightarrow x \in \left[-2, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 2\right]$$

$$\text{Hence, } D_f = \left[-2, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 2\right]$$

20. Given, $\sin^{-1} x + \sin^{-1} y = \pi$

It is possible only when each term of the given equation provides the maximum value.

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2} \text{ and } \sin^{-1} y = \frac{\pi}{2}$$

$$\Rightarrow x = \sin\left(\frac{\pi}{2}\right) = 1 \text{ and } y = \sin\left(\frac{\pi}{2}\right) = 1$$

Hence, the solutions are $x = 1$ and $y = 1$.

21. Given $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

It is possible only when each term will provide us the maximum value.

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2}, \sin^{-1} y = \frac{\pi}{2}$$

$$\text{and } \sin^{-1} z = \frac{\pi}{2}$$

$$\Rightarrow x = 1, y = 1 \text{ and } z = 1$$

Hence, the value of

$$\begin{aligned} & x^{2013} + y^{2013} + z^{2013} - \frac{9}{x^{2014} + y^{2014} + z^{2014}} \\ &= 1 + 1 + 1 - \frac{9}{1 + 1 + 1} \\ &= 3 - 3 \\ &= 0 \end{aligned}$$

22. We have $-\frac{\pi}{2} \leq \sin^{-1}(3x+5) \leq \frac{\pi}{2}$

$$\Rightarrow -\pi \leq 2 \sin^{-1}(3x+5) \leq \pi$$

$$\Rightarrow -\pi + \frac{\pi}{4} \leq 2 \sin^{-1}(3x+5) + \frac{\pi}{4} \leq \pi + \frac{\pi}{4}$$

$$\Rightarrow -\frac{3\pi}{4} \leq f(x) \leq \frac{5\pi}{4}$$

$$\text{Hence, } R_f = \left[-\frac{3\pi}{4}, \frac{5\pi}{4}\right]$$

23. We have $\sin^{-1} x > \sin^{-1} (3x - 1)$

$$\Rightarrow x > (3x - 1)$$

$$\Rightarrow 2x - 1 < 0$$

$$\Rightarrow x < \frac{1}{2}$$

$$\Rightarrow x \in \left[-1, \frac{1}{2}\right)$$

24. We have $-1 \leq 2x + 4 \leq 1$

$$\Rightarrow -5 \leq 2x \leq -3$$

$$\Rightarrow -\frac{5}{2} \leq x \leq -\frac{3}{2}$$

$$\text{Hence, } D_f = \left[-\frac{5}{2}, -\frac{3}{2}\right]$$

25. We have $0 \leq \cos^{-1} (3x + 5) \leq \pi$

$$\Rightarrow 0 \leq 2 \cos^{-1} (3x + 5) \leq 2\pi$$

$$\Rightarrow \frac{\pi}{4} \leq 2 \cos^{-1} (3x + 5) + \frac{\pi}{4} \leq 2\pi + \frac{\pi}{4}$$

$$\text{Hence, } R_f = \left[\frac{\pi}{4}, \frac{9\pi}{4}\right]$$

26. We have $\frac{\pi}{2} \leq \cos^{-1} (-x^2) \leq \pi$

$$\Rightarrow \frac{3\pi}{2} \leq 3 \cos^{-1} (-x^2) \leq 3\pi$$

$$\Rightarrow \frac{3\pi}{2} - \frac{\pi}{2} \leq 3 \cos^{-1} (-x^2) - \frac{\pi}{2} \leq 3\pi - \frac{\pi}{2}$$

$$\Rightarrow \pi \leq f(x) \leq \frac{5\pi}{2}$$

$$\text{Hence, } R_f = \left[\pi, \frac{5\pi}{2}\right]$$

27. Given, $\cos^{-1} x + \cos^{-1} x^2 = 0$

It is possible only when each term will provide us the minimum value.

$$\text{So, } \cos^{-1} x + \cos^{-1} x^2 = 0$$

$$\Rightarrow x = 1 \text{ and } x^2 = 1$$

$$\Rightarrow x = 1 \text{ and } x = \pm 1$$

Hence, the solution is $x = 1$.

28. Given, $[\sin^{-1} x] + [\cos^{-1} x]$ and $x \geq 0$

$$\Rightarrow [\sin^{-1} x] = 0 \text{ and } [\cos^{-1} x] = 0$$

$$\Rightarrow x \in [0, \sin 1] \text{ \& } x \in (\cos 1, 1]$$

$$\Rightarrow x \in (\cos 1, \sin 1)$$

29. We have $-1 \leq \frac{x^2}{x^2 + 1} \leq 1$

$$\Rightarrow \left| \frac{x^2}{x^2 + 1} \right| \leq 1$$

$$\Rightarrow \frac{|x^2|}{|x^2 + 1|} \leq 1$$

$$\Rightarrow \frac{x^2}{x^2 + 1} \leq 1$$

$$\Rightarrow x^2 + 1 \geq x^2$$

$$\Rightarrow 1 > 0$$

Hence, $x \in R$

30. We have $\cos^{-1} (x) > \cos^{-1} (x^2)$

$$\Rightarrow x < x^2$$

$$\Rightarrow x^2 - x > 0$$

$$\Rightarrow x(x - 1) > 0$$

$$\Rightarrow x \in [-1, 0)$$

31. Since $\tan^{-1} x$ is defined for all real values of x , so

$$9 - x^2 \leq 0$$

$$\Rightarrow x^2 - 9 \leq 0$$

$$\Rightarrow (x + 3)(x - 3) \leq 0$$

$$\Rightarrow -3 \leq x \leq 3$$

$$\text{Hence, } D_f = [-3, 3]$$

32. We have $-\frac{\pi}{2} \leq \tan^{-1} (1 - x^2) \leq \frac{\pi}{4}$

$$\Rightarrow -\pi \leq 2 \tan^{-1} (1 - x^2) \leq \frac{\pi}{2}$$

$$\Rightarrow -\pi + \frac{\pi}{6} \leq 2 \tan^{-1} (1 - x^2) + \frac{\pi}{6} \leq \frac{\pi}{2} + \frac{\pi}{6}$$

$$\Rightarrow -\frac{5\pi}{6} \leq f(x) \leq \frac{2\pi}{3}$$

$$\text{Hence, } R_f = \left[-\frac{5\pi}{6}, \frac{2\pi}{3}\right]$$

33. We have $f(x) = \cot^{-1} (2x - x^2)$

$$\Rightarrow f(x) = \cot^{-1} (1 - (x^2 - 2x + 1))$$

$$\Rightarrow f(x) = \cot^{-1} (1 - (x - 1)^2)$$

Since $(1 - (x - 1)^2) \leq 1$ and $0 \leq \cot^{-1} x \leq \pi$ and $\cot^{-1} x$ is strictly decreasing function so,

$$\cot^{-1} (1) \leq \cot^{-1} (1 - (x - 1)^2) \leq \cot^{-1} (0)$$

$$\Rightarrow \frac{\pi}{4} \leq f(x) \leq \pi$$

$$\text{Hence, } R_f = \left[\frac{\pi}{4}, \pi\right]$$

where $[\cdot]$ = GIF

34. We have $[\cot^{-1} x] + [\cos^{-1} x] = 0$

$$\Rightarrow [\cot^{-1} x] = 0 \text{ \& } [\cos^{-1} x] = 0$$

$$\Rightarrow 0 \leq \cot^{-1} x < 1 \text{ \& } 0 \leq \cos^{-1} x < 1$$

$$\Rightarrow x \in (\cot 1, \infty) \text{ \& } x \in (\cos 1, 1]$$

$$\Rightarrow x \in (\cot 1, 1]$$

35. We have $\sin \{x\} = \cos \{x\}$, $\forall x \in [0, 2\pi]$

$$\Rightarrow \tan \{x\} = 1$$

$$\Rightarrow \{x\} = \tan^{-1} (1) = \frac{\pi}{4}$$

Hence, the number of solutions = 6

(Since $\{x\}$ is a periodic function with period 1, it has one solution between 0 to 1. So, there is six solutions between 0 to 6.28).

36. We have,

$$-1 \leq \left(\frac{|x| - 2}{3} \right) \leq 1$$

$$\begin{aligned} \Rightarrow -3 \leq |x| - 2 \leq 3 \\ \Rightarrow -1 \leq |x| \leq 5 \\ \Rightarrow -5 \leq x \leq 5 \end{aligned} \quad \dots(i)$$

Also, $1 \leq \left(\frac{1-|x|}{4} \right) \leq 1$

$$\begin{aligned} \Rightarrow -4 \leq 1 - |x| \leq 4 \\ \Rightarrow -4 \leq |x| - 1 \leq 4 \\ \Rightarrow -3 \leq |x| \leq 5 \\ \Rightarrow -5 \leq x \leq 5 \end{aligned} \quad \dots(ii)$$

From (i) and (ii), we get
 $-5 \leq x \leq 5$

Thus, $D_f = [-5, 5]$

37. Given, $f(x) = \sin^{-1}(2x^2 - 1)$

So, $-1 \leq (2x^2 - 1) \leq 1$

$$\Rightarrow 0 \leq 2x^2 \leq 2$$

$$\Rightarrow 0 \leq x^2 \leq 1$$

$$\Rightarrow 0 \leq |x| \leq 1$$

$$\Rightarrow -1 \leq x \leq 1$$

Thus, $D_f = [-1, 1]$

38. Given, $f(x) = \sqrt{5\pi \sin^{-1}x - 6(\sin^{-1}x)^2}$

We have $5\pi \sin^{-1}x - 6(\sin^{-1}x)^2 \geq 0$

$$\Rightarrow (5\pi - 6(\sin^{-1}x)) \sin^{-1}x \geq 0$$

$$\Rightarrow (6(\sin^{-1}x) - 5\pi) \sin^{-1}x \leq 0$$

$$\Rightarrow 0 \leq \sin^{-1}x \leq \frac{5\pi}{6}$$

$$\Rightarrow 0 \leq x \leq \sin\left(\frac{5\pi}{6}\right)$$

$$\Rightarrow 0 \leq x \leq \frac{1}{2}$$

Also, $-1 \leq x \leq 1$

Thus, $D_f = \left[0, \frac{1}{2}\right]$

39. Given, $f(x) = \log_2 \left(\frac{3 \tan^{-1}x + \pi}{\pi - 4 \tan^{-1}x} \right)$

So, $\frac{3 \tan^{-1}x + \pi}{\pi - 4 \tan^{-1}x} > 0$

$$\Rightarrow \frac{3 \tan^{-1}x + \pi}{4 \tan^{-1}x - \pi} < 0$$

$$\Rightarrow \frac{\pi}{3} < \tan^{-1}x < \frac{\pi}{4}$$

$$\Rightarrow \tan\left(-\frac{\pi}{3}\right) < x < \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \sqrt{3} < x < 1$$

Hence, $D_f = (\sqrt{3}, 1)$

40. Given, $f(x) = \cos^{-1}\left(\frac{3}{2 + \sin x}\right)$

$$\Rightarrow 1 \leq \left(\frac{3}{2 + \sin x} \right) \leq 1$$

$$\Rightarrow -1 \leq \left(\frac{2 + \sin x}{3} \right) \leq 1$$

$$\Rightarrow -3 \leq (2 + \sin x) \leq 3$$

$$\Rightarrow -5 \leq (\sin x) \leq 1$$

$$\Rightarrow -1 \leq (\sin x) \leq 1$$

$$\Rightarrow \sin^{-1}(-1) \leq x \leq \sin^{-1}(1)$$

$$\Rightarrow \frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

Thus, $D_f = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

41. Given, $f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$

$$\Rightarrow 1 \leq \left(\frac{x^2 + 1}{2x} \right) \leq 1$$

$$\Rightarrow \left| \frac{x^2 + 1}{2x} \right| \leq 1$$

$$\Rightarrow \frac{x^2 + 1}{2|x|} \leq 1$$

$$\Rightarrow x^2 + 1 \leq 2|x|$$

$$\Rightarrow |x|^2 + 1 \leq 2|x|$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow (|x| - 1) \leq 0$$

$$\Rightarrow (|x| - 1) = 0$$

$$\Rightarrow x = \pm 1$$

Thus, $D_f = \{-1, 1\}$

42. Given, $f(x) = \cos^{-1}\left(\frac{x^2 + 1}{x^2}\right)$

So, $1 \leq \left(\frac{x^2 + 1}{x^2} \right) \leq 1$

$$\Rightarrow 1 \leq \left(1 + \frac{1}{x^2} \right) \leq 1$$

$$\Rightarrow 2 \leq \left(\frac{1}{x^2} \right) \leq 0$$

which is not true

Hence, $D_f = \emptyset$

43. Given, $f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$

So, $-1 \leq \log_2(x^2 + 3x + 4) \leq 1$

$$\Rightarrow \frac{1}{2} \leq (x^2 + 3x + 4) \leq 2$$

when $(x^2 + 3x + 4) \leq 2$

$$\Rightarrow (x^2 + 3x + 2) \leq 0$$

$$\Rightarrow (x + 1)(x + 2) \leq 0$$

$$\Rightarrow -2 \leq x \leq -1$$

when $x^2 + 3x + 4 \geq \frac{1}{2}$

$$\Rightarrow 2x^2 + 6x + 7 \geq 0$$

Clearly, $D < 0$

So, it is true for all R

Hence, $D_f = [-2, -1]$

44. Given, $f(x) = \sin^{-1}\left(\log_2\left(\frac{x^2}{2}\right)\right)$

So, $1 \leq \log_2\left(\frac{x^2}{2}\right) \leq 1$

$\Rightarrow \frac{1}{2} \leq \left(\frac{x^2}{2}\right) \leq 2$

$\Rightarrow 1 \leq x^2 \leq 4$

$\Rightarrow 1 \leq |x| \leq 2$... (i)

when $|x| \leq 2$

$\Rightarrow -2 \leq x \leq 2$

when $|x| \geq 1$

$\Rightarrow x \geq 1$ and $x \leq -1$... (ii)

From (i) and (ii), we get,

$x \in [-2, -1] \cup [1, 2]$

Thus, $D_f = [-2, -1] \cup [1, 2]$

45. Given, $f(x) = \sin^{-1}[2 - 3x^2]$

$\Rightarrow 1 \leq [2 - 3x^2] \leq 1$

when $[2 - 3x^2] \geq 1$

$\Rightarrow 1 \leq 2 - 3x^2 < 2$

$\Rightarrow 2 < 3x^2 - 2 \leq 1$

$\Rightarrow 0 < 3x^2 \leq 3$

$\Rightarrow 0 < x^2 \leq 1$

$\Rightarrow 0 < |x| \leq 1$

$\Rightarrow -1 \leq x \leq 1 - \{0\}$... (i)

Also, when $[2 - x^2] \geq 1$

$\Rightarrow 2 - x^2 \geq 1$

$\Rightarrow x^2 \leq 3$

$\Rightarrow |x| \leq \sqrt{3}$

$\Rightarrow \sqrt{3} \leq x \leq \sqrt{3}$... (ii)

Hence, $D_f = [1, 0) \cup (0, 1]$

46. Given, $f(x) = \frac{1}{x} + 3^{\sin^{-1}x} + \frac{1}{\sqrt{x-2}}$

Let $D_1 = R - \{0\}$

$D_2 = [1, 1]$

and $D_3 = (2, \infty)$

Thus, $D_f = D_1 \cap D_2 \cap D_3 = [1, 1]$

47. Given, $f(x) = \sin^{-1}(\log_2 x^2)$

So, $1 \leq \log_2(x^2) \leq 1$

$\Rightarrow \frac{1}{2} \leq (x^2) \leq 2$

$\Rightarrow \frac{1}{\sqrt{2}} \leq |x| \leq \sqrt{2}$

Thus, $x \in \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \sqrt{2}\right]$

Hence, $D_f = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \sqrt{2}\right]$

48. Given, $f(x) = e^x + \sin^{-1}\left(\frac{x}{2} - 1\right) + \frac{1}{x}$

Let $D_1 = R$

$D_2: 1 \leq \left(\frac{x}{2} - 1\right) \leq 1$

$\Rightarrow 0 \leq \left(\frac{x}{2}\right) \leq 2$

$\Rightarrow 0 \leq x \leq 4$

and $D_3 = R - \{0\}$

Hence, $D_f = D_1 \cap D_2 \cap D_3 = (0, 4]$

49. Given, $f(x) = \sqrt{\sin^{-1}(\log_x 2)}$

We have $\sin^{-1}(\log_x 2) \geq 0$

$\Rightarrow (\log_x 2) \geq \sin(0) = 0$

$\Rightarrow 2 \geq x^0 = 1$

$\Rightarrow 2 > 1$

which is true for all x in R

Also, $x \neq 1$ and $x > 0$

Furthermore, $1 \leq \log_x 2 \leq 1$ which is also true for $x \neq 1$ and $x > 0$

Hence, $D_f = (0, 1) \cup (1, \infty)$

50. Given, $f(x) = \sqrt{\sin^{-1}(\log_2 x)}$

We have $\sin^{-1}(\log_2 x) \geq 0$

$\Rightarrow (\log_2 x) \geq 0$

$\Rightarrow x \geq 1$

Also, $1 \leq \log_2 x \leq 1$

$\Rightarrow \frac{1}{2} \leq x \leq 2$

Hence, $D_f = [1, 2]$

51. Given, $f(x) = \sin^{-1}(2x - 3)$

$R_f = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

52. Given, $f(x) = 2 \sin^{-1}(2x - 1) \cdot \frac{\pi}{4}$

$-\frac{\pi}{2} \leq \sin^{-1}(2x - 1) \leq \frac{\pi}{2}$

$\pi \leq 2 \sin^{-1}(2x - 1) \leq \pi$

$\pi \cdot \frac{\pi}{4} \leq 2 \sin^{-1}(2x - 1) \cdot \frac{\pi}{4} \leq \pi \cdot \frac{\pi}{4}$

$\frac{5\pi}{4} \leq \left(2 \sin^{-1}(2x - 1) \cdot \frac{\pi}{4}\right) \leq \frac{3\pi}{4}$

$-\frac{5\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$

So, $R_f = \left[\frac{5\pi}{4}, \frac{3\pi}{4}\right]$

53. Given, $f(x) = 2 \cos^{-1}(x^2) - \pi$

$= 2(\pi - \cos^{-1}(x^2)) - \pi$

$= \pi - 2 \cos^{-1}(x^2)$

54. Given, $f(x) = \frac{1}{2} \tan^{-1}(1 - x^2) \cdot \frac{\pi}{4}$

Now, $-\infty < (1 - x^2) \leq 1$

$\Rightarrow \tan^{-1}(-\infty) < \tan^{-1}(1 - x^2) \leq \tan^{-1}(1)$

$\Rightarrow \frac{\pi}{2} < \tan^{-1}(1 - x^2) \leq \frac{\pi}{4}$

$$\Rightarrow -\frac{\pi}{4} < \frac{1}{2} \tan^{-1}(1-x^2) \leq \frac{\pi}{8}$$

$$\Rightarrow -\frac{\pi}{4} - \frac{\pi}{4} < \frac{1}{2} \tan^{-1}(1-x^2) - \frac{\pi}{4} \leq \frac{\pi}{8} - \frac{\pi}{4}$$

$$\Rightarrow -\frac{\pi}{2} < f(x) \leq -\frac{\pi}{8}$$

$$\text{So, } R_f = \left[-\frac{\pi}{2}, -\frac{\pi}{8} \right]$$

$$55. \text{ Given, } f(x) = \cot^{-1}(2x-x^2) \\ = \cot^{-1}(1-(x-1)^2)$$

$$\text{Clearly, } -\infty < (1-(x-1)^2) \leq 1$$

$$\Rightarrow \cot^{-1}(1) \leq \cot^{-1}((1-(x-1)^2)) \leq \cot^{-1}(-\infty)$$

$$\Rightarrow \frac{\pi}{4} \leq \cot^{-1}((1-(x-1)^2)) \leq \pi$$

$$\Rightarrow \frac{\pi}{4} \leq f(x) \leq \pi$$

$$\text{So, } R_f = \left[\frac{\pi}{4}, \pi \right]$$

$$56. \text{ Given, } f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$$

$$D_f = [1, 1]$$

$$\text{So, } R_f = [f(1), f(1)]$$

$$= \left[\frac{\pi}{2} + \frac{\pi}{4} + \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4} \right]$$

$$= \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$$

$$57. \text{ Given, } f(x) = \sin^{-1}x + \sec^{-1}x + \tan^{-1}x$$

$$\text{Thus, } D_f = \{1, 1\}$$

$$\text{So, } R_f = \{f(1), f(1)\}$$

$$= \left\{ \frac{\pi}{2} + 0 - \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4} \right\}$$

$$= \left\{ \frac{3\pi}{4}, \frac{3\pi}{4} \right\}$$

$$58. \text{ Given, } f(x) = 3\cot^{-1}x + 2\tan^{-1}x + \frac{\pi}{4} \\ = 2(\tan^{-1}x + \cot^{-1}x) + \cot^{-1}x + \frac{\pi}{4}$$

$$= 2 \times \frac{\pi}{2} + \cot^{-1}x + \frac{\pi}{4}$$

$$= \cot^{-1}x + \frac{5\pi}{4}$$

$$\text{Thus, } 0 \leq \cot^{-1}x \leq \pi$$

$$\Rightarrow 0 + \frac{5\pi}{4} \leq \cot^{-1}x + \frac{5\pi}{4} \leq \pi + \frac{5\pi}{4}$$

$$\Rightarrow \frac{5\pi}{4} \leq f(x) \leq \frac{9\pi}{4}$$

$$\text{So, } R_f = \left[\frac{5\pi}{4}, \frac{9\pi}{4} \right]$$

$$59. \text{ Given, } f(x) = \operatorname{cosec}^{-1}[1 + \sin^2 x].$$

$$\text{Clearly,}$$

$$1 \leq (1 + \sin^2 x) \leq 2$$

$$R_f = [\operatorname{cosec}^{-1}(2), \operatorname{cosec}^{-1}(1)]$$

$$= \left[\sin^{-1}\left(\frac{1}{2}\right), \sin^{-1}(1) \right]$$

$$= \left[\frac{\pi}{6}, \frac{\pi}{2} \right]$$

$$60. \text{ Given, } f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$$

$$\text{Clearly, } D_f = [2, 1]$$

$$\text{Thus, } R_f = [f(2), f(1)]$$

$$= \left[\frac{\pi}{2}, \frac{\pi}{2} \right] = \left\{ \frac{\pi}{2} \right\}$$

$$61. \text{ We have, } f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x \text{ is defined only when } 1 \leq x \leq 1$$

$$\text{Now, } f(1) = \sin^{-1}(1) + \cos^{-1}(1) + \tan^{-1}(1)$$

$$= \frac{\pi}{2} + 0 + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\text{and } f(1) = \sin^{-1}(1) + \cos^{-1}(-1) + \tan^{-1}(1)$$

$$= \frac{\pi}{2} + \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\text{Thus, } R_f = \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$$

$$62. \text{ We have } 4 \sin^{-1}(x-1) + \cos^{-1}(x-2) = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-2) + \frac{\pi}{2} = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-2) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}(x-2) = \frac{\pi}{6}$$

$$\Rightarrow (x-2) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\Rightarrow x = 2 + \frac{1}{2} = \frac{5}{2}$$

$$\text{Hence, the solution is } x = \frac{5}{2}$$

$$63. \text{ As we know that, if } \sin^{-1}(f(x)) + \cos^{-1}(g(x)) = \frac{\pi}{2}, \text{ then}$$

$$f(x) = g(x)$$

$$\Rightarrow (x^2 - 2x + 1) = (x^2 - x)$$

$$\Rightarrow 2x - x = 1$$

$$\Rightarrow x = 1$$

$$\text{Hence, the solution is } x = 1$$

$$64. \text{ We have}$$

$$\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$$

$$\Rightarrow \cos^{-1}\left(\frac{1}{\sqrt{x^2+x+1}}\right) + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$$

$$\Rightarrow \left(\frac{1}{\sqrt{x^2+x+1}}\right) = \sqrt{x^2+x+1}$$

$$\Rightarrow x^2+x+1 = 1$$

$$\Rightarrow x^2+x = 0$$

$$\Rightarrow x(x+1) = 0$$

$$\Rightarrow x = 0 \text{ and } -1$$

Hence, the number of solutions is 2

65. As we know that, if $\sin^{-1}(f(x)) + \cos^{-1}(g(x)) = \frac{\pi}{2}$, then

$$f(x) = g(x)$$

$$\Rightarrow \left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \right) = \left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \right)$$

$$\Rightarrow x \left(1 - \frac{x}{2} + \frac{x^2}{4} - \dots \right) = x^2 \left(1 - \frac{x^2}{2} + \frac{x^4}{4} - \dots \right)$$

$$\Rightarrow x \left(\frac{1}{1 + \frac{x}{2}} \right) = x^2 \left(\frac{1}{1 + \frac{x^2}{2}} \right)$$

$$\Rightarrow \left(\frac{2x}{x+2} \right) = \left(\frac{2x^2}{x^2+2} \right)$$

$$\Rightarrow x \left\{ \left(\frac{1}{x+2} \right) - \left(\frac{x}{x^2+2} \right) \right\} = 0$$

$$\Rightarrow x = 0 \text{ and } \left(\frac{1}{x+2} \right) = \left(\frac{x}{x^2+2} \right)$$

$$\Rightarrow x = \text{and } x = 1$$

66. We have $\sin^{-1} x > \cos^{-1} x$

$$\Rightarrow 2 \sin^{-1} x > \sin^{-1} x + \cos^{-1} x$$

$$\Rightarrow 2 \sin^{-1} x > \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x > \frac{\pi}{4}$$

$$\Rightarrow x > \sin \left(\frac{\pi}{4} \right)$$

$$\Rightarrow x > \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \in \left(\frac{1}{\sqrt{2}}, 1 \right]$$

67. $(\sin^{-1} x)^2 - 3 \sin^{-1} x + 2 = 0$

$$\Rightarrow (\sin^{-1} x - 1)(\sin^{-1} x - 2) = 0$$

$$\Rightarrow (\sin^{-1} x - 1) = 0, (\sin^{-1} x - 2) = 0$$

$$\Rightarrow \sin^{-1} x = 1, 2$$

$$\Rightarrow \sin^{-1} x = 1$$

$$\Rightarrow x \sin(1)$$

68. Given equation is $\sin^{-1} x + \sin^{-1} 2y = \pi$. It is possible only when

$$\Rightarrow \sin^{-1} x = \frac{\pi}{2}, \sin^{-1}(2y) = \frac{\pi}{2}$$

$$\Rightarrow x = 1, 2y = 1$$

$$\Rightarrow x = 1, y = \frac{1}{2}$$

69. Given equation is $\cos^{-1} x + \cos^{-1} x^2 = 2\pi$. It is possible only when

$$\Rightarrow \cos^{-1} x = \pi, \cos^{-1}(x^2) = \pi$$

$$\Rightarrow x = -1, x^2 = -1$$

$$\Rightarrow x = \varnothing$$

70. Given equation is $\cos^{-1} x + \cos^{-1} x^2 = 0$

It is possible only when

$$\Rightarrow \cos^{-1} x = 0, \cos^{-1}(x^2) = 0$$

$$\Rightarrow x = 1 \text{ and } x^2 = 1$$

$$\Rightarrow x = 1$$

71. Given equation is

$$4 \sin^{-1}(x-1) + \cos^{-1}(x-1) = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-1) + \frac{\pi}{2} = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-1) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}(x-1) = \frac{\pi}{6}$$

$$\Rightarrow (x-1) = \sin \left(\frac{\pi}{6} \right) = \frac{1}{2}$$

$$\Rightarrow x = \frac{3}{2}$$

Hence, the solution is $x = \frac{3}{2}$

72. Given equation is

$$\cot^{-1} \left(\frac{1}{x^2-1} \right) + \tan^{-1}(x^2-1) = \frac{\pi}{2}$$

It is possible only when

$$\Rightarrow \frac{1}{x^2-1} = x^2-1$$

$$\Rightarrow (x^2-1)^2 = 1$$

$$\Rightarrow (x^2-1) = \pm 1$$

$$\Rightarrow x^2 = 1 \pm 1 = 2, 0$$

$$\Rightarrow x = \{-\sqrt{2}, 0, \sqrt{2}\}$$

73. Given equation is

$$\cot^{-1} \left(\frac{x^2-1}{2x} \right) + \tan^{-1} \left(\frac{2x}{x^2-1} \right) = \frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1} \left(\frac{2x}{x^2-1} \right) + \tan^{-1} \left(\frac{2x}{x^2-1} \right) = \frac{2\pi}{3}$$

$$\Rightarrow 2 \tan^{-1} \left(\frac{2x}{x^2-1} \right) = \frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1} \left(\frac{2x}{x^2-1} \right) = \frac{\pi}{3}$$

$$\Rightarrow \tan^{-1} \left(\frac{2x}{1-x^2} \right) = -\frac{\pi}{3}$$

$$\Rightarrow 2 \tan^{-1} x = -\frac{\pi}{3}$$

$$\Rightarrow \tan^{-1} x = -\frac{\pi}{6}$$

$$\Rightarrow x = \tan \left(-\frac{\pi}{6} \right) = -\frac{1}{\sqrt{3}}$$

Hence, the solution is $x = -\frac{1}{\sqrt{3}}$.

74. Given equation is

$$4 \sin^{-1} x + \cos^{-1} x = \frac{3\pi}{4}$$

$$\Rightarrow 3 \sin^{-1} x + \frac{\pi}{2} = \frac{3\pi}{4}$$

$$\Rightarrow 3 \sin^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow \sin^{-1} x = \frac{\pi}{12}$$

$$\Rightarrow x = \sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

75. Given equation is

$$5 \tan^{-1} x + 3 \cot^{-1} x = \frac{7\pi}{4}$$

$$\Rightarrow 2 \tan^{-1} x + \frac{3\pi}{2} = \frac{7\pi}{4}$$

$$\Rightarrow 2 \tan^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} x = \frac{\pi}{8}$$

$$\Rightarrow x = \tan\left(\frac{\pi}{8}\right) = (\sqrt{2}-1)$$

76. Given equation is

$$5 \tan^{-1} x + 4 \cot^{-1} x = 2\pi$$

$$\Rightarrow \tan^{-1} x + 2\pi = 2\pi$$

$$\Rightarrow \tan^{-1} x = 0$$

$$\Rightarrow x = \tan(0) = 0$$

Hence, the solution is $x = 0$.

77. Given equation is

$$\cot^{-1} x - \cot^{-1}(x+1) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{1}{x+1}\right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{x} - \frac{1}{x+1}}{1 + \frac{1}{x} \times \frac{1}{x+1}}\right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{x^2+x+1}\right) = \frac{\pi}{2}$$

$$\Rightarrow \left(\frac{1}{x^2+x+1}\right) = \tan\left(\frac{\pi}{2}\right) = \infty$$

$$\Rightarrow x^2+x+1 = \frac{1}{\infty} = 0$$

$$\Rightarrow x^2+x+1 = 0$$

So, no real values of x satisfies the above equation.

Hence, the solution is $x = \phi$

78. Given equation is

$$[\sin^{-1} x] + [\cos^{-1} x] = 0$$

It is possible only when

$$[\sin^{-1} x] = 0, [\cos^{-1} x] = 0$$

$$\Rightarrow 0 \leq \sin^{-1} x \leq 1 \text{ and } 0 \leq \cos^{-1} x \leq 1$$

$$\Rightarrow 0 \leq x \leq \sin(1) \text{ and } \cos(1) \leq x \leq 1$$

$$\Rightarrow x \in [\cos(1), \sin(1)]$$

79. Given equation is

$$[\tan^{-1} x] + [\cot^{-1} x] = 0$$

It is possible only when

$$[\tan^{-1} x] = 0 \text{ and } [\cot^{-1} x] = 0$$

$$\Rightarrow 0 \leq \tan^{-1} x \leq 1 \text{ and } 0 \leq \cot^{-1} x \leq 1$$

$$\Rightarrow \cot(1) \leq x \leq \tan(1)$$

Hence, $x \in [\cot(1), \tan(1)]$

80. Given equation is

$$[\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 0$$

$$\Rightarrow 0 \leq \sin^{-1}(\cos^{-1}(\sin^{-1}(\tan^{-1} x))) < 1$$

$$\Rightarrow 0 \leq (\cos^{-1}(\sin^{-1}(\tan^{-1} x))) < \sin(1)$$

$$\Rightarrow \cos(\sin(1)) < (\sin^{-1}(\tan^{-1} x)) \leq 1$$

$$\Rightarrow \sin(\cos(\sin(1))) < (\tan^{-1} x) \leq \sin(1)$$

$$\Rightarrow \tan(\sin(\cos(\sin(1)))) < x \leq \tan(\sin(1))$$

81. Do yourself.

82. Given equation is

$$(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$$

$$\Rightarrow (\tan^{-1} x + \cot^{-1} x)^2 - 2 \tan^{-1} x \cdot \cot^{-1} x = \frac{5\pi^2}{8}$$

$$\Rightarrow \frac{\pi^2}{4} - 2a\left(\frac{\pi}{2} - a\right) = \frac{5\pi^2}{8}, a = \tan^{-1} x$$

$$\Rightarrow 2a\left(\frac{\pi}{2} - a\right) + \frac{3\pi^2}{8} = 0$$

$$\Rightarrow a\pi - 2a^2 + \frac{3\pi^2}{8} = 0$$

$$\Rightarrow 8a\pi - 16a^2 + 3\pi^2 = 0$$

$$\Rightarrow 16a^2 - 8a\pi - 3\pi^2 = 0$$

$$\Rightarrow 16a^2 - 12a\pi + 4a\pi - 3\pi^2 = 0$$

$$\Rightarrow 4a(4a - 3\pi) + \pi(4a - 3\pi) = 0$$

$$\Rightarrow (4a + \pi)(4a - 3\pi) = 0$$

$$\Rightarrow a = \frac{3\pi}{4}, -\frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} x = \frac{3\pi}{4}, -\frac{\pi}{4}$$

$$\Rightarrow x = \tan\left(\frac{3\pi}{4}\right), \tan\left(-\frac{\pi}{4}\right)$$

$$x = -1$$

$$83. \text{ Let } \frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\Rightarrow \cos^{-1}\left(\frac{3}{5}\right) = 2\theta$$

$$\Rightarrow \cos(2\theta) = \frac{3}{5}$$

$$\Rightarrow 2 \cos^2 \theta - 1 = \frac{3}{5}$$

$$\Rightarrow 2 \cos^2 \theta = 1 + \frac{3}{5} = \frac{8}{5}$$

$$\Rightarrow \cos^2 \theta = \frac{4}{5}$$

$$\Rightarrow \cos \theta = \frac{2}{\sqrt{5}}$$

84. We have

$$\begin{aligned} & \sin\left(\frac{\pi}{4} + \sin^{-1}\left(\frac{1}{2}\right)\right) \\ &= \sin\left(\frac{\pi}{4} + \theta\right), \theta = \sin^{-1}\left(\frac{1}{2}\right) \\ &= \sin\left(\frac{\pi}{4} + \theta\right), \sin \theta = \frac{1}{2} \\ &= \sin\left(\frac{\pi}{4}\right) \cos(\theta) + \cos\left(\frac{\pi}{4}\right) \sin(\theta) \\ &= \frac{1}{\sqrt{2}} \cos(\theta) + \frac{1}{\sqrt{2}} \sin(\theta) \\ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \end{aligned}$$

85. Let m_1 and m_2 be the roots of $x^2 + 3x + 1 = 0$

$$\text{Thus, } m_1 + m_2 = -3 < 0$$

$$\text{and } m_1 \cdot m_2 = 1$$

It is possible only when both are negative.

$$\begin{aligned} \text{Thus, } \tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right) &= \tan^{-1}(m) - \pi + \cot^{-1}(m) \\ &= \tan^{-1}(m) + \cot^{-1}(m) - \pi \\ &= \frac{\pi}{2} - \pi \\ &= -\frac{\pi}{2} \end{aligned}$$

86. We have $\cos(\tan^{-1}(\sin(\cot^{-1} x)))$

$$= \cos(\tan^{-1}(\sin \theta)), \cot \theta = x$$

$$= \cos\left(\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)$$

$$= \cos \phi, \tan \phi = \left(\frac{1}{\sqrt{1+x^2}}\right)$$

$$= \sqrt{\frac{x^2+1}{x^2+2}}$$

87. Given, $6(\sin^{-1} x)^2 - \pi \sin^{-1} x \leq 0$

$$\Rightarrow \sin^{-1} x (6 \sin^{-1} x - \pi) \leq 0$$

$$\Rightarrow 0 \leq \sin^{-1} x \leq \frac{\pi}{6}$$

$$\Rightarrow 0 \leq x \leq \frac{1}{2}$$

88. Given, in-equation is

$$\frac{2 \tan^{-1} x + \pi}{4 \tan^{-1} x - \pi} \leq 0$$

$$\Rightarrow -\frac{\pi}{2} \leq \tan^{-1} x \leq \frac{\pi}{4}$$

$$\Rightarrow -\infty < x < 1$$

89. Given inequation is

$$\Rightarrow \sin^{-1} x < \sin^{-1} x^2$$

$$\Rightarrow x^2 > x$$

$$\Rightarrow x(x-1) > 0$$

$$\Rightarrow x > 1 \text{ and } x < 0$$

$$\Rightarrow x \in [-1, 0)$$

90. Given in-equation is

$$\Rightarrow \cos^{-1} x > \cos^{-1} x^2$$

$$\Rightarrow x^2 > x$$

$$\Rightarrow x^2 - x > 0$$

$$\Rightarrow x(x-1) > 0$$

$$\Rightarrow -1 \leq x < 0$$

91. Given in-equation is

$$\log_2(\tan^{-1} x) > 1$$

$$\Rightarrow \tan^{-1} x > 2$$

$$\Rightarrow x > \tan(2)$$

Hence, the solution is

$$(\tan 2, \infty)$$

92. Given in-equation is

$$(\cot^{-1} x)^2 - 5 \cot^{-1} x + 6 > 0$$

$$\Rightarrow (\cot^{-1} x - 2)(\cot^{-1} x - 3) > 0$$

$$\Rightarrow (\cot^{-1} x - 2) < 0, (\cot^{-1} x - 3) > 0$$

$$\Rightarrow x > \cot(2), x < \cot(3)$$

$$\Rightarrow x \in (\cot 2, \cot 3)$$

93. Given in-equation is

$$\sin^{-1} x < \cos^{-1} x$$

$$\Rightarrow 2 \sin^{-1} x < \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x < \frac{\pi}{4}$$

$$\Rightarrow x < \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \in \left[-1, \frac{1}{\sqrt{2}}\right)$$

94. Given in-equation is

$$\sin^{-1} x > \sin^{-1}(1-x)$$

$$x > (1-x)$$

$$2x > 1$$

$$x > \frac{1}{2}$$

$$\text{Hence, the solution is } x \in \left(\frac{1}{2}, 1\right]$$

95. Given in-equation is

$$\sin^{-1} 2x > \operatorname{cosec}^{-1} x$$

$$\Rightarrow \sin^{-1}(2x) > \sin^{-1}\left(\frac{1}{x}\right)$$

$$\Rightarrow 2x > \frac{1}{x}$$

$$\begin{aligned} \Rightarrow 2x \cdot \frac{1}{x} &> 0 \\ \Rightarrow \frac{2x^2 - 1}{x} &> 0 \\ \Rightarrow \frac{(\sqrt{2}x + 1)(\sqrt{2}x - 1)}{x} &> 0 \\ \Rightarrow x \in \left(-\frac{1}{\sqrt{2}}, 0\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right] \end{aligned}$$

96. Given in-equation is

$$\begin{aligned} \tan^{-1} 3x &< \cot^{-1} x \\ \Rightarrow \tan^{-1}(3x) &< \tan^{-1}\left(\frac{1}{x}\right) \\ \Rightarrow (3x) \left(\frac{1}{x}\right) &< 0 \\ \Rightarrow \frac{(\sqrt{3}x + 1)(\sqrt{3}x - 1)}{x} &< 0 \\ \Rightarrow x \in \left(-\frac{1}{\sqrt{3}}, 0\right) \cup \left(0, \frac{1}{\sqrt{3}}\right) \end{aligned}$$

97. Given in-equation is

$$\begin{aligned} \cos^{-1} 2x &\geq \sin^{-1} x \\ \Rightarrow \sin^{-1} \sqrt{1 - 4x^2} &> \sin^{-1} x \\ \Rightarrow \sqrt{1 - 4x^2} &> x \\ \Rightarrow (1 - 4x^2) &> x^2 \\ \Rightarrow 5x^2 - 1 &< 0 \\ \Rightarrow (\sqrt{5}x + 1)(\sqrt{5}x - 1) &< 0 \\ \Rightarrow -\frac{1}{\sqrt{5}} < x < \frac{1}{\sqrt{5}} \end{aligned}$$

98. Given in-equation is

$$\begin{aligned} x^2 - 2x &< \sin^{-1}(\sin 2) \\ \Rightarrow x^2 - 2x &< (\pi - 2) \\ \Rightarrow (x - 1)^2 &< (\pi - 1) \\ \Rightarrow |(x - 1)| &< \sqrt{(\pi - 1)} \\ \Rightarrow -\sqrt{(\pi - 1)} &< (x - 1) < \sqrt{(\pi - 1)} \\ \Rightarrow 1 - \sqrt{(\pi - 1)} &< x < 1 + \sqrt{(\pi - 1)} \end{aligned}$$

99. Given in-equation is

$$\begin{aligned} \sin^{-1}\left(\frac{x}{2}\right) &< \cos^{-1}(x + 1) \\ \Rightarrow \sin^{-1}\left(\frac{x}{2}\right) &< \sin^{-1}(\sqrt{1 - (x + 1)^2}) \\ \Rightarrow \left(\frac{x}{2}\right) &< (\sqrt{1 - (x + 1)^2}) \\ \Rightarrow \left(\frac{x}{2}\right)^2 &< 1 - (x + 1)^2 \\ \Rightarrow \frac{x^2}{4} &< -2x - x^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{5x^2}{4} + 2x &< 0 \\ \Rightarrow 5x^2 + 8x &< 0 \\ \Rightarrow x(5x + 8) &< 0 \\ \Rightarrow -\frac{8}{5} < x < 0 \end{aligned}$$

100. Given in-equation is

$$\begin{aligned} \tan^{-1} 2x &> 2 \tan^{-1} x \\ \Rightarrow \tan^{-1}(2x) &> \tan^{-1}\left(\frac{2x}{1 - x^2}\right) \\ \Rightarrow (2x) &> \left(\frac{2x}{1 - x^2}\right) \\ \Rightarrow (2x) \left(1 - \frac{1}{1 - x^2}\right) &> 0 \\ \Rightarrow x \left(\frac{1 - x^2 - 1}{1 - x^2}\right) &> 0 \\ \Rightarrow x \left(\frac{x^2}{x^2 - 1}\right) &> 0 \\ \Rightarrow \left(\frac{x^3}{(x - 1)(x + 1)}\right) &> 0 \\ \Rightarrow x \in (-1, 0) \cup (1, \infty) \end{aligned}$$

101. Given in-equation is

$$\begin{aligned} \tan(\cos^{-1} x) &\leq \sin\left(\cot^{-1}\left(\frac{1}{2}\right)\right) \\ \Rightarrow \frac{\sqrt{1 - x^2}}{x} &\leq \frac{1}{\sqrt{5}} \\ \Rightarrow \left(\frac{1 - x^2}{x^2}\right) &\leq \frac{1}{5} \\ \Rightarrow \left(\frac{1 - x^2}{x^2} - \frac{1}{5}\right) &\leq 0 \\ \Rightarrow \frac{5 - 5x^2 - x^2}{5x^2} &\leq 0 \\ \Rightarrow \frac{6x^2 - 5}{x^2} &\leq 0 \\ \Rightarrow \frac{6x^2 - 5}{x^2} &\geq 0 \\ \Rightarrow x \in \mathbb{R} - \left(-\sqrt{\frac{5}{6}}, \sqrt{\frac{5}{6}}\right) \end{aligned}$$

102. As we know that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, for every x in $[-1, 1]$

(i) Since $0 < \frac{1}{m^2 + 1} \leq 1$

$$\text{so, } f\left(\frac{1}{m^2 + 1}\right) = \frac{\pi}{2}$$

(ii) Since, $0 \leq \frac{m^2}{m^2+1} < 1$,

$$\text{so } f\left(\frac{m^2}{m^2+1}\right) = \frac{\pi}{2}.$$

(iii) Since, $-\frac{1}{2} \leq \frac{m}{m^2+1} \leq \frac{1}{2}$

$$\text{so, } f\left(\frac{m}{m^2+1}\right) = \frac{\pi}{2}$$

(iv) Since $m^2 - 2m + 6 = (m-1)^2 + 5$

$$\text{Thus, } 4 \leq (m-1)^2 + 5 < \infty$$

Hence, $f((m-1)^2 + 5)$ is not defined.

(v) Also, $1 \leq (m-1)^2 + 5 < \infty$

So, $f(m^2 + 1)$ is not defined.

103. Given, $\cos^{-1}x + \cos^{-1}y = \frac{2\pi}{3}$

$$\text{Now, } \sin^{-1}x + \sin^{-1}y$$

$$= \frac{\pi}{2} \cos^{-1}x + \frac{\pi}{2} \cos^{-1}y$$

$$= \pi (\cos^{-1}x + \cos^{-1}y)$$

$$= \pi \frac{2\pi}{3}$$

$$= \frac{\pi}{3}$$

104. Let m_1 and m_2 be the two roots of the given equation.

$$\text{Now, } m_1 + m_2 = 3 \text{ and } m_1 \cdot m_2 = 1$$

$$\Rightarrow m_1 \text{ and } m_2 \text{ are two negative roots.}$$

$$\text{Now, } \tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right)$$

$$= \tan^{-1}(m) - \pi + \cot^{-1}(m)$$

$$= -\pi + \tan^{-1}(m) + \cot^{-1}(m)$$

$$= \pi + \frac{\pi}{2}$$

$$= -\frac{\pi}{2}$$

105. Let $m = \frac{2x^2+5}{x^2+2} = 2 + \frac{1}{x^2+2}$

$$\text{Thus, } m \in \left[2, \frac{5}{2}\right]$$

$$\text{now, } \sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+2}\right)\right) > \sin^{-1}(\sin 3)$$

$$\Rightarrow \sin^{-1}\left(\sin\left(\pi - \frac{2x^2+5}{x^2+2}\right)\right)$$

$$< \sin^{-1}(\sin(\pi - 3))$$

$$\Rightarrow \pi - \left(\frac{2x^2+5}{x^2+2}\right) > \pi - 3$$

$$\Rightarrow \left(\frac{2x^2+5}{x^2+2}\right) < 3$$

$$\Rightarrow \left(\frac{2x^2+5}{x^2+2} - 3\right) < 0$$

$$\Rightarrow \left(\frac{x^2-5}{x^2+2}\right) < 0$$

$$\Rightarrow \left(\frac{x^2+5}{x^2+2}\right) > 0$$

$$\Rightarrow x \in R$$

106. (i) $\sin^{-1}(\sin 3)$

$$= \sin^{-1}(\sin(\pi - 3))$$

$$= (\pi - 3)$$

(ii) $\sin^{-1}(\sin 5)$

$$= \sin^{-1}(\sin(5 - 2\pi))$$

$$= (5 - 2\pi)$$

(iii) $\sin^{-1}(\sin 7)$

$$= \sin^{-1}(\sin(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(iv) $\sin^{-1}(\sin 10)$

$$= \sin^{-1}(\sin(3\pi - 10))$$

$$= (3\pi - 10)$$

(v) $\sin^{-1}(\sin 20)$

$$= \sin^{-1}(\sin(20 - 6\pi))$$

$$= (20 - 6\pi)$$

107. (i) $\cos^{-1}(\cos 2) = 2$

(ii) $\cos^{-1}(\cos 3) = 3$

(iii) $\cos^{-1}(\cos 5)$

$$= \cos^{-1}(\cos(2\pi - 5))$$

$$= (2\pi - 5)$$

(iv) $\cos^{-1}(\cos 7)$

$$= \cos^{-1}(\cos(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(v) $\cos^{-1}(\cos 10)$

$$= \cos^{-1}(\cos(4\pi - 10))$$

$$= (4\pi - 10)$$

108. (i) $\tan^{-1}(\tan 3)$

$$= \tan^{-1}(\tan(3 - \pi))$$

$$= (3 - \pi)$$

(ii) $\tan^{-1}(\tan 5)$

$$= \tan^{-1}(\tan(5 - 2\pi))$$

$$= (5 - 2\pi)$$

(iii) $\tan^{-1}(\tan 7)$

$$= \tan^{-1}(\tan(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(iv) $\tan^{-1}(\tan 10)$

$$= \tan^{-1}(\tan(10 - 3\pi))$$

$$= (10 - 3\pi)$$

(v) $\tan^{-1}(\tan 15)$

$$= \tan^{-1}(\tan(15 - 5\pi))$$

$$= (15 - 5\pi)$$

109. We have

$$\cos^{-1}(\sin(-5))$$

$$= \cos^{-1}(\sin 5)$$

$$= \pi - \cos^{-1}(\sin 5)$$

$$\begin{aligned}
 &= \pi - \cos^{-1} \left(\cos \left(\frac{\pi}{2} - 5 \right) \right) \\
 &= \pi - \left(\frac{\pi}{2} - 5 \right) \\
 &= \frac{\pi}{2} + 5
 \end{aligned}$$

110. We have

$$\begin{aligned}
 f(x) &= \sin^{-1}(\sin x) \\
 &= x + 2\pi - \pi - x + x + \pi - x \\
 &= 2\pi
 \end{aligned}$$

$$\Rightarrow f(x) = 0$$

111. We have

$$\begin{aligned}
 f(x) &= \cos^{-1}(\cos x) \\
 &= x + x + 2\pi = x \\
 &= 2\pi - x
 \end{aligned}$$

$$\Rightarrow f(x) = -1$$

112. We have

$$\begin{aligned}
 &\sin^{-1} \left(\sin \left(\frac{2x^2 + 5}{x^2 + 1} \right) \right) < \pi - 3 \\
 \Rightarrow &\sin^{-1} \left(\sin \left(\pi - \left(\frac{2x^2 + 5}{x^2 + 1} \right) \right) \right) > \pi - 3 \\
 \Rightarrow &\left(\pi - \left(\frac{2x^2 + 5}{x^2 + 1} \right) \right) < \pi - 3 \\
 \Rightarrow &-\left(\frac{2x^2 + 5}{x^2 + 1} \right) < -3 \\
 \Rightarrow &\left(\frac{2x^2 + 5}{x^2 + 1} \right) > 3 \\
 \Rightarrow &\left(\frac{2x^2 + 5}{x^2 + 1} - 3 \right) > 0 \\
 \Rightarrow &\left(\frac{2x^2 + 5 - 3x^2 - 3}{x^2 + 1} \right) > 0 \\
 \Rightarrow &x^2 < 2 \\
 \Rightarrow &-\sqrt{2} < x < \sqrt{2}
 \end{aligned}$$

113. We have

$$\begin{aligned}
 &x^2 - 3x < \sin^{-1}(\sin 2) \\
 \Rightarrow &x^2 - 3x < \sin^{-1}(\sin(\pi - 2)) \\
 \Rightarrow &x^2 - 3x < (\pi - 2) \\
 \Rightarrow &x^2 - 3x + (2 - \pi) < 0 \\
 \Rightarrow &\left(x - \frac{3 + \sqrt{1 + 4\pi}}{2} \right) \left(x - \frac{3 - \sqrt{1 + 4\pi}}{2} \right) < 0 \\
 \Rightarrow &\frac{3 - \sqrt{1 + 4\pi}}{2} < x < \frac{3 + \sqrt{1 + 4\pi}}{2}
 \end{aligned}$$

114. We have

$$\begin{aligned}
 &= \sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) + \tan^{-1}(\tan 50) \\
 &= \sin^{-1}(\sin(50 - 16\pi)) + \cos^{-1}(\cos(16\pi - 50)) \\
 &\quad + \tan^{-1}(\tan(50 - 16\pi)) \\
 &= (50 - 16\pi) + (16\pi - 50) + (50 - 16\pi) \\
 &= (50 - 16\pi)
 \end{aligned}$$

$$\begin{aligned}
 115. \quad &\sin^{-1}(\sin 1) + \sin^{-1}(\sin 2) + \sin^{-1}(\sin 3) \\
 &= 1 + (\pi - 2) + (\pi - 3) \\
 &= (2\pi - 4)
 \end{aligned}$$

$$\begin{aligned}
 116. \quad &\sin^{-1}(\sin 10) + \sin^{-1}(\sin 20) \\
 &\quad + \sin^{-1}(\sin 30) + \sin^{-1}(\sin 40) \\
 &= (3\pi - 10) + (20 - 6\pi) + (30 - 10\pi) + (13\pi - 40) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 117. \quad &\cos^{-1}(\cos 1) + \cos^{-1}(\cos 2) + \cos^{-1}(\cos 3) + \cos^{-1}(\cos 4) \\
 &= 1 + 2 + 3 + (2\pi - 4) \\
 &= 2(\pi + 1)
 \end{aligned}$$

$$\begin{aligned}
 118. \quad &\cos^{-1}(\cos 10) + \cos^{-1}(\cos 20) \\
 &\quad + \cos^{-1}(\cos 30) + \cos^{-1}(\cos 40) \\
 &= (4\pi + 10) + (20 + 6\pi) + (10\pi + 30) + (40 - 12\pi) \\
 &= (20 - 4\pi)
 \end{aligned}$$

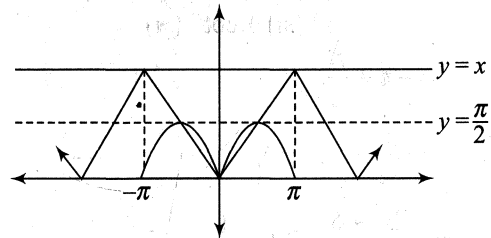
$$\begin{aligned}
 119. \quad &\sin^{-1}(\sin 10) + \cos^{-1}(\cos 10) \\
 &= (3\pi - 10) + (4\pi - 10) \\
 &= (7\pi - 20)
 \end{aligned}$$

$$\begin{aligned}
 120. \quad &\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) \\
 &= (50 - 16\pi) + (16\pi - 50) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 121. \quad &\sin^{-1}(\sin 100) + \cos^{-1}(\cos 100) \\
 &= (100 - 32\pi) + (32\pi - 100) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 122. \quad &\cos^{-1}(\sin(-5)) + \sin^{-1}(\cos(-5)) \\
 &= \pi - \cos^{-1}(\sin 5) + \sin^{-1}(\cos 5) \\
 &= \pi - \cos^{-1} \left(\cos \left(5 - \frac{3\pi}{2} \right) \right) + \sin^{-1} \left(\sin \left(5 - \frac{3\pi}{2} \right) \right) \\
 &= \pi - \left(5 - \frac{3\pi}{2} \right) + \left(5 - \frac{3\pi}{2} \right) \\
 &= \pi
 \end{aligned}$$

123.



Hence, the number of solutions is 3.

Thus, the ordered pairs are

$$\left(-\frac{\pi}{2}, 1 \right), (0, 0), \left(\frac{\pi}{2}, 1 \right)$$

124. Given,

$$\begin{aligned}
 f(x) &= \cos^{-1}(\cos x) - \sin^{-1}(\sin x) \\
 &= x - (x + \pi - x) \\
 &= x - \pi
 \end{aligned}$$

Hence, the required area

$$= \frac{1}{2} \times \pi \times \pi = \frac{\pi^2}{2}$$

$$\begin{aligned}
 125. \quad &\tan^{-1}(\tan 1) + \tan^{-1}(\tan 2) \\
 &\quad + \tan^{-1}(\tan 3) + \tan^{-1}(\tan 4) \\
 &= 1 + (2 - \pi) + (3 - \pi) + (4 - \pi) \\
 &= (10 - 3\pi)
 \end{aligned}$$

$$\begin{aligned}
 126. \quad & \tan^{-1}(\tan 20) + \tan^{-1}(\tan 40) \\
 & + \tan^{-1}(\tan 60) + \tan^{-1}(\tan 80) \\
 & = (20 - 6\pi) + (40 - 13\pi) + (60 - 21\pi) + (80 - 26\pi) \\
 & = (200 - 66\pi)
 \end{aligned}$$

$$\begin{aligned}
 127. \quad & \sin^{-1}(\sin 15) + \cos^{-1}(\cos 15) + \tan^{-1}(\tan 15) \\
 & = (5\pi - 15) + (15 - 4\pi) + (15 - 5\pi) \\
 & = (15 - 4\pi)
 \end{aligned}$$

$$\begin{aligned}
 128. \quad & \sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) - \tan^{-1}(\tan 50) \\
 & = (50 - 16\pi) + (16\pi - 50) - (50 - 16\pi) \\
 & = (16\pi - 50)
 \end{aligned}$$

$$\begin{aligned}
 129. \quad & 3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4) \\
 \Rightarrow & 3x^2 + 2x < 2(\pi - 4) - (2\pi - 4) \\
 \Rightarrow & 3x^2 + 2x < -4 \\
 \Rightarrow & 3x^2 + 8x + 4 < 0 \\
 \Rightarrow & 3x^2 + 6x + 2x + 4 < 0 \\
 \Rightarrow & 3x(x + 2) + 2(x + 2) < 0 \\
 \Rightarrow & (3x + 2)(x + 2) < 0 \\
 \Rightarrow & -2 < x < -\frac{2}{3}
 \end{aligned}$$

130. We have

$$\begin{aligned}
 & \sin^{-1}\left(\sin\left(\frac{2x^2 + 4}{x^2 + 1}\right)\right) < \pi - 3 \\
 \Rightarrow & \sin^{-1}\left(\sin\left(\pi - \left(\frac{2x^2 + 4}{x^2 + 1}\right)\right)\right) < \pi - 3 \\
 \Rightarrow & \left(\pi - \left(\frac{2x^2 + 4}{x^2 + 1}\right)\right) < \pi - 3 \\
 \Rightarrow & -\left(\frac{2x^2 + 4}{x^2 + 1}\right) < -3 \\
 \Rightarrow & \left(\frac{2x^2 + 4}{x^2 + 1}\right) > 3 \\
 \Rightarrow & \left(\frac{2x^2 + 4}{x^2 + 1} - 3\right) > 0 \\
 \Rightarrow & \left(\frac{2x^2 + 4 - 3x^2 - 3}{x^2 + 1}\right) > 0 \\
 \Rightarrow & x^2 < 1 \\
 \Rightarrow & -1 < x < 1
 \end{aligned}$$

$$\begin{aligned}
 131. \quad & \text{We have } \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) \\
 & = \tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}}\right) \\
 & = \tan^{-1}\left(\frac{5/6}{1 - 1/6}\right) \\
 & = \tan^{-1}\left(\frac{5/6}{5/6}\right) = \tan^{-1}(1) = \frac{\pi}{4}
 \end{aligned}$$

132. We have

$$\begin{aligned}
 & \tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) \\
 & = \frac{\pi}{4} + \pi + \tan^{-1}\left(\frac{2+3}{1-2 \cdot 3}\right) \\
 & = \frac{\pi}{4} + \pi + \tan^{-1}(-1) \\
 & = \frac{\pi}{4} + \pi - \frac{\pi}{4} \\
 & = \pi
 \end{aligned}$$

133. We have

$$\begin{aligned}
 & \tan^{-1}(9) + \tan^{-1}\left(\frac{5}{4}\right) \\
 & = \pi + \tan^{-1}\left(\frac{9 + \frac{5}{4}}{1 - 9 \cdot \frac{5}{4}}\right) \\
 & = \pi + \tan^{-1}\left(\frac{\frac{41}{4}}{\frac{-41}{4}}\right) \\
 & = \pi + \tan^{-1}(-1) \\
 & = \pi - \frac{\pi}{4} = \frac{3\pi}{4}
 \end{aligned}$$

134. We have

$$\begin{aligned}
 & \sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) - \sin^{-1}\left(\frac{63}{65}\right) \\
 & = \sin^{-1}\left(\frac{4}{5} \cdot \sqrt{1 - \left(\frac{5}{13}\right)^2} + \frac{5}{13} \cdot \sqrt{1 - \left(\frac{4}{5}\right)^2}\right) - \sin^{-1}\left(\frac{63}{65}\right) \\
 & = \sin^{-1}\left(\frac{4}{5} \cdot \frac{12}{13} + \frac{5}{13} \cdot \frac{3}{5}\right) - \sin^{-1}\left(\frac{63}{65}\right) \\
 & = \sin^{-1}\left(\frac{48}{65} + \frac{15}{65}\right) - \sin^{-1}\left(\frac{63}{65}\right) \\
 & = \sin^{-1}\left(\frac{63}{65}\right) - \sin^{-1}\left(\frac{63}{65}\right) \\
 & = 0
 \end{aligned}$$

135. We have

$$\begin{aligned}
 & 2 \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) \\
 & = \tan^{-1}\left(\frac{2 \cdot \frac{1}{3}}{1 - \left(\frac{1}{3}\right)^2}\right) + \tan^{-1}\left(\frac{1}{7}\right) \\
 & = \tan^{-1}\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right) + \tan^{-1}\left(\frac{1}{7}\right) \\
 & = \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right)
 \end{aligned}$$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \cdot \frac{1}{7}} \right) \\
 &= \tan^{-1} \left(\frac{25}{25} \right) \\
 &= \frac{\pi}{4}
 \end{aligned}$$

136. Let $\sin^{-1} x = A$, $\sin^{-1} y = B$, $\sin^{-1} z = C$

Then $x = \sin A$, $y = \sin B$, $z = \sin C$

$$\begin{aligned}
 \text{we have, } &x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} \\
 &= \sin A \cdot \cos A + \sin B \cdot \cos B + \sin C \cdot \cos C \\
 &= \frac{1}{2}(\sin 2A + \sin 2B + \sin 2C) \\
 &= \frac{1}{2}(4\sin A \cdot \sin B \cdot \sin C) \\
 &= 2 \sin A \cdot \sin B \cdot \sin C \\
 &= 2xyz
 \end{aligned}$$

137. We have

$$\begin{aligned}
 &\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi \\
 \Rightarrow &\cos^{-1} x + \cos^{-1} y = \pi - \cos^{-1} z \\
 \Rightarrow &\cos^{-1} x + \cos^{-1} y = \cos^{-1}(-z) \\
 \Rightarrow &\cos^{-1}(xy \sqrt{1-x^2} \sqrt{1-y^2}) = \cos^{-1}(-z) \\
 \Rightarrow &(xy \sqrt{1-x^2} \sqrt{1-y^2})^2 = (-z)^2 \\
 \Rightarrow &x^2 y^2 + 2xyz + z^2 = 1 - x^2 - y^2 + x^2 y^2 \\
 \Rightarrow &x^2 + y^2 + z^2 + 2xyz = 1
 \end{aligned}$$

138. Given, $\cos^{-1} \left(\frac{x}{2} \right) + \cos^{-1} \left(\frac{y}{3} \right) = \theta$

$$\begin{aligned}
 \Rightarrow &\cos^{-1} \left(\frac{x}{2} \cdot \frac{y}{3} - \sqrt{1 - \frac{x^2}{4}} \sqrt{1 - \frac{y^2}{9}} \right) = \theta \\
 \Rightarrow &\left(\frac{xy}{6} - \sqrt{1 - \frac{x^2}{4}} \sqrt{1 - \frac{y^2}{9}} \right) = \cos \theta \\
 \Rightarrow &\left(\frac{xy}{6} - \cos \theta \right)^2 = \left(1 - \frac{x^2}{4} \right) \left(1 - \frac{y^2}{9} \right) \\
 &\frac{x^2 y^2}{36} - \frac{xy}{3} \cos \theta + \cos^2 \theta \\
 \Rightarrow &= 1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2 y^2}{36} \\
 \Rightarrow &\frac{x^2}{4} + \frac{y^2}{9} - \frac{xy}{3} \cos \theta = 1 - \cos^2 \theta \\
 \Rightarrow &\frac{x^2}{4} + \frac{y^2}{9} - \frac{xy}{3} \cos \theta = \sin^2 \theta \\
 \Rightarrow &9x^2 + 4y^2 - 12xy \cos \theta = 36 \sin^2 \theta
 \end{aligned}$$

139. We have

$$\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) = \pi$$

and $\cot^{-1}(1) + \cot^{-1}(2) + \cot^{-1}(3)$

$$\begin{aligned}
 &= \tan^{-1}(1) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) \\
 &= \frac{\pi}{4} + \frac{\pi}{4} \\
 &= \frac{\pi}{2}
 \end{aligned}$$

Hence, $m = \frac{\pi}{\pi/2} = 2$

140. **Case I:** When $x \leq 0$

Then, $\tan^{-1}(2x) \leq 0$, $\tan^{-1}(3x) \leq 0$,

$\Rightarrow x \leq 0$

So, it has no solution.

Case II: When $x > 0$, $2x \cdot 3x = 6x^2 < 1$

$$\Rightarrow x < \frac{1}{\sqrt{6}}$$

$$\text{Then } \frac{3\pi}{4} = \tan^{-1}(2x) + \tan^{-1}(3x) < \frac{\pi}{2}$$

So, it is not possible.

Case III: When $x > 0$, $2x \cdot 3x > 1$

$$\Rightarrow x > \frac{1}{\sqrt{6}}$$

$$\text{Then } \frac{3\pi}{4} = \tan^{-1}(2x) + \tan^{-1}(3x)$$

$$\Rightarrow \pi + \tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{3\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{3\pi}{4} \quad \pi = \frac{\pi}{4}$$

$$\Rightarrow \left(\frac{5x}{1-6x^2}\right) = 1$$

$$\Rightarrow 6x^2 - 5x + 1 = 0$$

$$\Rightarrow x = 1, \frac{1}{6}$$

Thus, $x = 1$ is a solution.

141. We have

$$\sin^{-1}(x) + \sin^{-1}(2x) = \frac{\pi}{3}$$

$$\Rightarrow \sin^{-1} x + \sin^{-1}(2x) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow \sin^{-1} x - \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \sin^{-1}(2x)$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{2} - \frac{\sqrt{3}}{2}\sqrt{1-x^2}\right) = \sin^{-1}(2x)$$

$$\Rightarrow \left(\frac{x}{2} - \frac{\sqrt{3}}{2}\sqrt{1-x^2}\right) = 2x$$

$$\Rightarrow 5x = \sqrt{3}\sqrt{1-x^2}$$

$$\Rightarrow 25x^2 = 3(1-x^2)$$

$$\Rightarrow 28x^2 = 3$$

$$\Rightarrow x = \pm \frac{\sqrt{3}}{2\sqrt{7}}$$

$$\Rightarrow x = \frac{\sqrt{3}}{2\sqrt{7}}, \text{ negative value of } x \text{ does not satisfy the given equation.}$$

142. We have

$$\begin{aligned} f(x) &= \cos^{-1}(x) + \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right) \\ &= \cos^{-1}(x) + \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}(x) \\ &= \frac{\pi}{3} \end{aligned}$$

$$\text{Now, } f(2013) = \frac{\pi}{3}$$

143. $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$

$$\Rightarrow \sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1}x$$

$$\Rightarrow (1-x) = \sin\left(\frac{\pi}{2} - 2\sin^{-1}x\right)$$

$$\Rightarrow (1-x) = \cos(2\sin^{-1}x)$$

$$\Rightarrow (1-x) = 1 - 2x^2$$

$$\Rightarrow x(2x-1) = 0$$

$$\Rightarrow x = 0, (2x-1) = 0$$

$$\Rightarrow x = 0, \frac{1}{2}$$

144. $x^2 - 4x > \sin^{-1}(\sin[\pi^{3/2}]) + \cos^{-1}(\cos[\pi^{3/2}])$

$$\Rightarrow x^2 - 4x > \sin^{-1}(\sin 5.5) + \cos^{-1}(\cos 5.5)$$

$$\Rightarrow x^2 - 4x > (5.5 - 2\pi) + (2\pi - 5.5)$$

$$\Rightarrow x^2 - 4x > 0$$

$$\Rightarrow x(x-4) > 0$$

$$\Rightarrow x < 0 \text{ and } x > 4$$

$$\Rightarrow x \in (-\infty, 0) \cup (4, \infty)$$

145. $\cos(\tan^{-1}x) = x$

$$\Rightarrow \frac{1}{\sqrt{x^2+1}} = x$$

$$\Rightarrow x^2(x^2+1) = 1$$

$$\Rightarrow x^4 + x^2 - 1 = 0$$

$$\Rightarrow x^2 = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Rightarrow x^2 = \frac{\sqrt{5}-1}{2}$$

$$\Rightarrow x = \pm \sqrt{\frac{\sqrt{5}-1}{2}}$$

146. $\sin(\tan^{-1}x) = \cos(\cot^{-1}(x+1))$

$$\Rightarrow \frac{x}{\sqrt{x^2+1}} = \frac{x+1}{\sqrt{1+(x+1)^2}}$$

$$\Rightarrow \frac{x^2}{x^2+1} = \frac{(x+1)^2}{1+(x+1)^2}$$

$$\Rightarrow \frac{x^2}{x^2+1} \cdot 1 = \frac{(x+1)^2}{1+(x+1)^2} \cdot 1$$

$$\Rightarrow \frac{-1}{x^2+1} = \frac{-1}{1+(x+1)^2}$$

$$\Rightarrow x^2+1 = 1+(x+1)^2$$

$$\Rightarrow x^2 = (x+1)^2$$

$$\Rightarrow 2x = -1$$

$$\Rightarrow x = -\frac{1}{2}$$

147. $\sec^{-1}\left(\frac{x}{2}\right) + \sec^{-1}x = \sec^{-1}2$

$$\Rightarrow \cos^{-1}\left(\frac{x}{2}\right) = \cos^{-1}\left(\frac{1}{x}\right) + \cos^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow \cos^{-1}\left(\frac{x}{2}\right) = \cos^{-1}\left(\frac{1}{2} \cdot \frac{1}{x} - \sqrt{1-\frac{1}{x^2}} \cdot \frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow \left(\frac{x}{2}\right) = \left(\frac{1}{2} \cdot \frac{1}{x} - \sqrt{1-\frac{1}{x^2}} \cdot \frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow x - \frac{1}{x} = -\sqrt{3} \sqrt{1-\frac{1}{x^2}}$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 3\left(1 - \frac{1}{x^2}\right)$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 3 - \frac{3}{x^2}$$

$$\Rightarrow x^2 + \frac{4}{x^2} - 5 = 0$$

$$\Rightarrow x^4 - 5x^2 + 4 = 0$$

$$\Rightarrow (x^2-1)(x^2-4) = 0$$

$$\Rightarrow x = \pm 2, \pm 1$$

148. $\cos\left(\tan^{-1}\left(\cot\left(\sin^{-1}\left(x + \frac{3}{2}\right)\right)\right)\right) + \tan(\sec^{-1}x) = 0$

$$\Rightarrow \left(\frac{2x+3}{2}\right) = \sqrt{x^2-1}$$

$$\Rightarrow (2x+3)^2 = 4(x^2-1)$$

$$\Rightarrow 12x+9 = -4$$

$$\Rightarrow x = -\frac{13}{12}$$

149. Given equation is

$$\tan\left(\tan^{-1}\left(\frac{x}{10}\right) + \tan^{-1}\left(\frac{1}{x+1}\right)\right) = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{x}{10} + \frac{1}{x+1}}{1 - \frac{x}{10} \cdot \frac{1}{x+1}}\right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{x^2+x+12}{11x+12} = 1$$

$$\Rightarrow x^2+x+12 = 11x+12$$

$$\Rightarrow x^2 - 10x = 0$$

$$\Rightarrow x(x-10) = 0$$

$$\Rightarrow x = 0, 10$$

$$150. (x-2)x^2 + 8x + k + 4 \\ > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$$

$$\Rightarrow (k-2)x^2 + 8x + k + 4 \\ > (12-4\pi) + (4\pi-12) \\ \Rightarrow (k-2)x^2 + 8x + (k+4) > 0$$

For all x in R , $D \geq 0$

$$\Rightarrow 64 - 4(k-2)(k+4) \geq 0$$

$$\Rightarrow 16 - (k-2)(k+4) \geq 0$$

$$\Rightarrow (k-2)(k+4) - 16 \leq 0$$

$$\Rightarrow k^2 + 2k - 24 \leq 0$$

$$\Rightarrow (k+6)(k-4) \leq 0$$

$$\Rightarrow -6 \leq k \leq 4$$

Thus, the least integral value of k is -6

151. Do yourself.

152. Given,

$$f(x) = \sin^{-1}(\sin x), \forall x \in [-\pi, 2\pi]. \\ = (-\pi - x) - x + (\pi - x) + (x - 2\pi) \\ = -2\pi - 2x$$

Thus, $f'(x) = -2$

153. Given,

$$f(x) = \cos^{-1}(\cos x), \forall x \in [-2\pi, \pi] \\ = (x + 2\pi) - x + x \\ = (x + 2\pi)$$

Thus, $f'(x) = 1$

154. Given,

$$f(x) = \tan^{-1}(\tan x), \forall x \in \left[-\frac{3\pi}{2}, \frac{5\pi}{2}\right] \\ = (x + \pi) + x + (x - \pi) + (x - 2\pi) \\ = (4x - 2\pi)$$

Thus, $f'(x) = 4$

155. We have

$$\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cot^{-1}(3) \\ = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) \\ = \tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}}\right) \\ = \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$156. \text{ We have } 2 \tan^{-1}\left(\frac{3}{2}\right) + \tan^{-1}\left(\frac{12}{5}\right)$$

$$= \tan^{-1}\left(\frac{2 \cdot \frac{3}{2}}{1 - \frac{9}{4}}\right) + \tan^{-1}\left(\frac{12}{5}\right)$$

$$= \tan^{-1}\left(-\frac{12}{5}\right) + \tan^{-1}\left(\frac{12}{5}\right)$$

$$= \pi - \tan^{-1}\left(\frac{12}{5}\right) + \tan^{-1}\left(\frac{12}{5}\right)$$

$$= \pi$$

$$157. \cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right) \\ = \cot^{-1}\left(\frac{\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right) - \cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)}\right) \\ = \cot^{-1}\left(\frac{\cos\left(\frac{x}{2}\right)}{\sin\left(\frac{x}{2}\right)}\right) \\ = \cot^{-1}\left(\cot\left(\frac{x}{2}\right)\right) = \frac{x}{2}$$

$$158. \sin^{-1}(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}) \\ = \sin^{-1}(x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2}) \\ = \sin^{-1}(x) - \sin^{-1}(\sqrt{x})$$

$$159. \sin^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right), -\frac{\pi}{4} < x < \frac{\pi}{4} \\ = \sin^{-1}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) \\ = \sin^{-1}\left(\sin\left(x + \frac{\pi}{4}\right)\right) \\ = \left(x + \frac{\pi}{4}\right)$$

$$160. \cos^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right), \frac{\pi}{4} < x < \frac{5\pi}{4} \\ = \cos^{-1}\left(\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x\right) \\ = \cos^{-1}\left(\cos\left(x - \frac{\pi}{4}\right)\right) \\ = \left(x - \frac{\pi}{4}\right)$$

$$161. \tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right)$$

Put $x^2 = \cos 2\theta$

$$= \tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}\right)$$

$$= \tan^{-1}\left(\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}\right)$$

$$= \tan^{-1}\left(\frac{1 + \tan \theta}{1 - \tan \theta}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right)$$

$$= \left(\frac{\pi}{4} + \theta\right)$$

$$= \left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1}(x^2) \right)$$

$$\begin{aligned} 162. \sin^{-1} \left(\frac{3}{5} \cos x + \frac{4}{5} \sin x \right) \\ &= \sin^{-1} (\sin \alpha \cos x + \cos \alpha \sin x) \\ &= \sin^{-1} (\sin (x + \alpha)) \\ &= (x + \alpha) \\ &= x + \tan^{-1} \left(\frac{3}{4} \right) \end{aligned}$$

$$163. \text{ Let } \sin^{-1} \left(\frac{1}{4} \right) = \theta$$

$$\Rightarrow \sin \theta = \frac{1}{4}$$

Now,

$$\begin{aligned} \sin(2\theta) &= 2 \sin \theta \cdot \cos \theta \\ &= 2 \times \frac{1}{4} \times \sqrt{1 - \frac{1}{16}} \\ &= \frac{\sqrt{15}}{8} \end{aligned}$$

$$164. \text{ Let } \cos^{-1} \left(\frac{1}{3} \right) = \theta$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

Now,

$$\begin{aligned} \cos(2\theta) &= 2 \cos^2 \theta - 1 \\ &= \frac{2}{9} - 1 \\ &= -\frac{7}{9} \end{aligned}$$

$$165. \text{ Let } \tan^{-1} \left(\frac{1}{3} \right) = \theta$$

$$\Rightarrow \tan \theta = \frac{1}{3}$$

Now,

$$\begin{aligned} \cos(2\theta) &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \\ &= \frac{1 - \left(\frac{1}{3} \right)^2}{1 + \left(\frac{1}{3} \right)^2} \\ &= \frac{9 - 1}{9 + 1} \\ &= \frac{8}{10} = \frac{4}{5} \end{aligned}$$

$$166. \text{ Let } \cot^{-1} \left(\frac{3}{4} \right) = \theta$$

$$\text{Now, } \sin \left(\frac{\theta}{2} \right)$$

$$\begin{aligned} &= \sqrt{\frac{1 - \cos \theta}{2}} \\ &= \sqrt{\frac{1 - \frac{\cos \theta}{\sin \theta \cdot \operatorname{cosec} \theta}}{2}} \\ &= \sqrt{\frac{1 - \frac{\cot \theta}{\operatorname{cosec} \theta}}{2}} \\ &= \sqrt{\frac{1 - \frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}}{2}} \\ &= \frac{1}{\sqrt{2}} \sqrt{1 - \frac{3/4}{\sqrt{1 + 9/16}}} \\ &= \frac{1}{\sqrt{2}} \sqrt{1 - \frac{3}{5}} \\ &= \frac{2}{\sqrt{5}} \end{aligned}$$

$$167. \text{ Let } \tan^{-1} \left(\frac{3}{4} \right) = \theta$$

$$\Rightarrow \tan \theta = \frac{3}{4}$$

We have

$$\begin{aligned} \tan \left(\frac{3\pi}{4} - 2\theta \right) &= \frac{\tan \left(\frac{3\pi}{4} \right) \tan(2\theta)}{1 + \tan \left(\frac{3\pi}{4} \right) \tan(2\theta)} \\ &= \frac{1 \tan(2\theta)}{1 + \tan(2\theta)} \\ &= \frac{-1 - \frac{2 \tan \theta}{1 - \tan^2 \theta}}{1 - \frac{2 \tan \theta}{1 - \tan^2 \theta}} \\ &= \frac{\tan^2 \theta - 2 \tan \theta - 1}{1 - \tan^2 \theta - 2 \tan \theta} \\ &= \frac{9 - 6 - 1}{16 - 9 - 6} \\ &= \frac{41}{17} \end{aligned}$$

$$168. \text{ Let } \sin^{-1} \left(\frac{1}{2} \right) = \theta$$

$$\text{Then } \sin \theta = \frac{1}{2}$$

$$\text{Now, } \sin \left(2 \sin^{-1} \left(\frac{1}{2} \right) \right)$$

$$\begin{aligned}
 &= \sin(2\theta) \\
 &= 2 \sin \theta \cos \theta \\
 &= 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}
 \end{aligned}$$

169. Let $\sin^{-1}\left(\frac{1}{3}\right) = \theta$

Then $\sin \theta = \frac{1}{3}$

Now, $\sin\left(3 \sin^{-1}\left(\frac{1}{3}\right)\right)$
 $= \sin(3\theta)$
 $= 3 \sin \theta - 4 \sin^3 \theta$
 $= 3 \cdot \frac{1}{3} - 4 \cdot \left(\frac{1}{3}\right)^3$
 $= 1 - \frac{4}{27} = \frac{23}{27}$

170. Let $\frac{1}{2} \cos^{-1}\left(\frac{1}{8}\right) = \theta$

Then $\cos^{-1}\left(\frac{1}{8}\right) = 2\theta$

$$\Rightarrow \cos(2\theta) = \frac{1}{8}$$

$$\Rightarrow 2 \cos^2 \theta - 1 = \frac{1}{8}$$

$$\Rightarrow 2 \cos^2 \theta = \frac{9}{8}$$

$$\Rightarrow \cos^2 \theta = \frac{9}{16}$$

$$\Rightarrow \cos \theta = \frac{3}{4}$$

$$\Rightarrow \cos\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{8}\right)\right) = \frac{3}{4}$$

171. Let $\frac{1}{2} \cos^{-1}\left(-\frac{1}{10}\right) = \theta$

Then $\cos^{-1}\left(-\frac{1}{10}\right) = 2\theta$

$$\Rightarrow \cos(2\theta) = -\frac{1}{10}$$

$$\Rightarrow 2 \cos^2 \theta - 1 = -\frac{1}{10}$$

$$\Rightarrow 2 \cos^2 \theta = 1 - \frac{1}{10} = \frac{9}{10}$$

$$\Rightarrow \cos^2 \theta = \frac{9}{20}$$

$$\Rightarrow \cos \theta = \frac{3}{2\sqrt{5}}$$

$$\Rightarrow \cos \theta = \frac{3}{2\sqrt{5}} = \frac{3\sqrt{5}}{2\sqrt{5} \times \sqrt{5}} = \frac{3\sqrt{5}}{10}$$

$$\Rightarrow \cos\left(\frac{1}{2} \cos^{-1}\left(-\frac{1}{10}\right)\right) = \frac{3\sqrt{5}}{10}$$

172. Let $\frac{1}{2} \cos^{-1}\left(\frac{1}{9}\right) = \theta$

$$\Rightarrow \cos^{-1}\left(\frac{1}{9}\right) = 2\theta$$

$$\Rightarrow \cos(2\theta) = \frac{1}{9}$$

$$\Rightarrow 2 \cos^2(\theta) - 1 = \frac{1}{9}$$

$$\Rightarrow 2 \cos^2(\theta) = 1 + \frac{1}{9} = \frac{10}{9}$$

$$\Rightarrow \cos^2(\theta) = \frac{5}{9}$$

$$\Rightarrow \sin^2(\theta) = 1 - \frac{5}{9} = \frac{4}{9}$$

$$\Rightarrow \sin(\theta) = \frac{2}{3}$$

$$\Rightarrow \sin\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{9}\right)\right) = \frac{2}{3}$$

173. Let $\frac{1}{4} \tan^{-1}(\sqrt{63}) = \theta$

$$\Rightarrow \tan^{-1}(\sqrt{63}) = 4\theta$$

$$\Rightarrow \tan(4\theta) = \sqrt{63}$$

$$\Rightarrow \tan(4\theta) = \sqrt{63}$$

$$\Rightarrow \frac{\sin(4\theta)}{\cos(4\theta)} = \sqrt{63}$$

$$\Rightarrow \frac{\sin(4\theta)}{\sqrt{63}} = \frac{\cos(4\theta)}{1} = \frac{1}{8}$$

Now, $\cos(4\theta) = \frac{1}{8}$

$$\Rightarrow 2 \cos^2(2\theta) - 1 = \frac{1}{8}$$

$$\Rightarrow 2 \cos^2(2\theta) = 1 + \frac{1}{8} = \frac{9}{8}$$

$$\Rightarrow \cos^2(2\theta) = \frac{9}{16}$$

$$\Rightarrow \cos(2\theta) = \frac{3}{4}$$

...(i)

$$\Rightarrow 2 \cos^2(\theta) - 1 = \frac{3}{4}$$

$$\Rightarrow 2 \cos^2(\theta) = \frac{7}{4}$$

$$\Rightarrow \cos^2(\theta) = \frac{7}{8}$$

$$\Rightarrow \cos(\theta) = \frac{\sqrt{7}}{2\sqrt{2}}$$

...(ii)

$$\text{Also, } \sin(4\theta) = \frac{\sqrt{63}}{8}$$

$$\Rightarrow 2 \sin(2\theta) \cos(2\theta) = \frac{\sqrt{63}}{8}$$

$$\Rightarrow 2 \sin(2\theta) \times \frac{3}{4} = \frac{\sqrt{63}}{8}, \text{ from (i)}$$

$$\Rightarrow \sin(2\theta) = \frac{\sqrt{63}}{12}$$

$$\Rightarrow 2 \sin(\theta) \cos(\theta) = \frac{\sqrt{63}}{12}, \text{ from (ii)}$$

$$\Rightarrow 2 \sin(\theta) \times \frac{\sqrt{7}}{2\sqrt{2}} = \frac{\sqrt{63}}{12}$$

$$\Rightarrow \sin(\theta) \times \frac{\sqrt{7}}{\sqrt{2}} = \frac{\sqrt{63}}{12}$$

$$\Rightarrow \sin(\theta) = \frac{3\sqrt{2}}{12}$$

$$\Rightarrow \sin(\theta) = \frac{\sqrt{2}}{4}$$

$$\Rightarrow \sin(\theta) = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin\left(\frac{1}{4} \tan^{-1}(\sqrt{63})\right) = \frac{1}{2\sqrt{2}}$$

$$174. \text{ Let } \frac{1}{4} \tan^{-1}\left(\frac{24}{7}\right) = \theta$$

$$\Rightarrow \tan^{-1}\left(\frac{24}{7}\right) = 4\theta$$

$$\Rightarrow \tan(4\theta) = \frac{24}{7}$$

$$\Rightarrow \frac{\sin(4\theta)}{24} = \frac{\cos(4\theta)}{7} = \frac{1}{25}$$

$$\text{Now, } \cos(4\theta) = \frac{7}{25}$$

$$\Rightarrow 2 \cos^2(2\theta) - 1 = \frac{7}{25}$$

$$\Rightarrow 2 \cos^2(2\theta) = 1 + \frac{7}{25} = \frac{32}{25}$$

$$\Rightarrow \cos^2(2\theta) = \frac{32}{50}$$

$$\Rightarrow \cos(2\theta) = \sqrt{\frac{32}{50}}$$

$$\Rightarrow 2 \cos^2(\theta) - 1 = \sqrt{\frac{32}{50}}$$

$$\Rightarrow 2 \cos^2(\theta) = 1 + \frac{8}{10} = \frac{18}{10}$$

$$\Rightarrow \cos^2(\theta) = \frac{9}{10}$$

$$\Rightarrow \cos(\theta) = \frac{3}{\sqrt{10}}$$

$$\Rightarrow \cos\left(\frac{1}{4} \tan^{-1}\left(\frac{24}{7}\right)\right) = \frac{3}{\sqrt{10}}$$

$$175. \text{ Let } \frac{1}{2} \cos^{-1}\left(\frac{2}{3}\right) = \theta$$

$$\Rightarrow \cos^{-1}\left(\frac{2}{3}\right) = 2\theta$$

$$\Rightarrow \cos(2\theta) = \frac{2}{3}$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{2}{3}$$

$$\Rightarrow 2 + 2 \tan^2 \theta = 3 - 3 \tan^2 \theta$$

$$\Rightarrow 5 \tan^2 \theta = 1$$

$$\Rightarrow \tan^2 \theta = \frac{1}{5}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \tan\left(\frac{1}{2} \cos^{-1}\left(\frac{2}{3}\right)\right) = \frac{1}{\sqrt{5}}$$

$$176. \text{ We have } \tan\left(2 \tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right) = -\frac{7}{17}$$

$$\text{Let } 2 \tan^{-1}\left(\frac{1}{5}\right) = \theta$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{2}{5}}{1 - \frac{1}{25}}\right) = \theta$$

$$\Rightarrow \tan \theta = \frac{\frac{2}{5}}{1 - \frac{1}{25}} = \frac{10}{24} = \frac{5}{12}$$

$$\text{Now, } \tan\left(\theta - \frac{\pi}{4}\right)$$

$$= \frac{\tan \theta - 1}{1 + \tan \theta}$$

$$= \frac{\frac{5}{12} - 1}{\frac{5}{12} + 1} = -\frac{7}{17}$$

$$177. \text{ Let } \frac{1}{4} \sin^{-1}\left(-\frac{4}{5}\right) = \theta$$

$$\Rightarrow \sin^{-1}\left(-\frac{4}{5}\right) = 4\theta$$

$$\Rightarrow \sin(4\theta) = -\frac{4}{5}$$

$$\Rightarrow \frac{2 \tan(2\theta)}{1 + \tan^2(2\theta)} = -\frac{4}{5}$$

$$\Rightarrow \frac{\tan(2\theta)}{1 + \tan^2(2\theta)} = -\frac{2}{5}$$

$$\Rightarrow 2 \tan^2(2\theta) + 5 \tan(2\theta) + 2 = 0$$

$$\Rightarrow \tan(2\theta) = -\frac{1}{2}, -2$$

$$\text{when } \tan(2\theta) = -\frac{1}{2}$$

$$\Rightarrow \frac{2 \tan \theta}{1 - \tan^2 \theta} = -\frac{1}{2}$$

$$\Rightarrow \tan^2 \theta - 4 \tan \theta - 1 = 0$$

$$\Rightarrow \tan \theta = 2 - \sqrt{5}$$

$$\text{Now, } \tan\left(\frac{3\pi}{4} - \frac{1}{4}\sin^{-1}\left(-\frac{4}{5}\right)\right)$$

$$= \tan\left(\frac{3\pi}{4} - \theta\right)$$

$$= \tan\left(\pi - \left(\frac{\pi}{4} + \theta\right)\right)$$

$$= -\tan\left(\frac{\pi}{4} + \theta\right)$$

$$= -\left(\frac{1 + \tan \theta}{1 - \tan \theta}\right)$$

$$= -\left(\frac{1 + 2 - \sqrt{5}}{1 - 2 + \sqrt{5}}\right)$$

$$= \left(\frac{3 - \sqrt{5}}{1 - \sqrt{5}}\right)$$

$$= \left(\frac{(3 - \sqrt{5})(1 + \sqrt{5})}{-4}\right)$$

$$= -\frac{1}{4}(3 + 2\sqrt{5} - 5)$$

$$= -\frac{1}{4}(-2 + 2\sqrt{5})$$

$$= \left(\frac{1 - \sqrt{5}}{2}\right)$$

178. Given equation is

$$x^2 - 3x < \sin^{-1}(\sin 2)$$

$$\Rightarrow x^2 - 3x < (\pi - 2)$$

$$\Rightarrow \left(x - \frac{3}{2}\right)^2 < \left(\pi - 2 + \frac{9}{4}\right) = \frac{4\pi + 1}{4}$$

$$\Rightarrow \left|x - \frac{3}{2}\right| < \frac{\sqrt{4\pi + 1}}{2}$$

$$\Rightarrow -\frac{\sqrt{4\pi + 1}}{2} < \left(x - \frac{3}{2}\right) < \frac{\sqrt{4\pi + 1}}{2}$$

$$\Rightarrow \frac{3}{2} - \frac{\sqrt{4\pi + 1}}{2} < x < \frac{\sqrt{4\pi + 1}}{2} + \frac{3}{2}$$

$$\Rightarrow -0.2 < x < 3.3$$

Thus, the integral values of x are 0, 1, 2, 3.

179. Given in-equation is

$$3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$$

$$\Rightarrow 3x^2 + 8x < 2(\pi - 4) - (2\pi - 4)$$

$$\Rightarrow 3x^2 + 8x + 4 < 0$$

$$\Rightarrow 3x^2 + 6x + 2x + 4 < 0$$

$$\Rightarrow 3x(x + 2) + 2(x + 2) < 0$$

$$\Rightarrow (3x + 2)(x + 2) < 0$$

$$\Rightarrow -2 < x < -\frac{2}{3}$$

180. We have

$$f(x) = \cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{1 - 3x^2}}{2}\right\}$$

$$= \cos^{-1}(x) + \cos^{-1}\left(\frac{1}{2}x + \sqrt{1 - x^2}\sqrt{1 - \frac{1}{4}}\right)$$

$$= \cos^{-1}(x) + \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}(x)$$

$$= \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3}$$

$$= \text{constant function.}$$

Hence, the result.

181. We have $f(x)$

$$= \sin^{-1}\left(\frac{2x}{1 + x^2}\right) + 2 \tan^{-1}(x)$$

$$= \pi - 2 \tan^{-1}(x) + 2 \tan^{-1}(x)$$

$$= \pi$$

Hence, the value of $f(2013) = \pi$

182. We have

$$f(x) = 2 \tan^{-1}\left(\frac{1 + x}{1 - x}\right) + \sin^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= 2(\tan^{-1}(1) + \tan^{-1}(x)) + \frac{\pi}{2} - \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= 2\left(\frac{\pi}{4} + \tan^{-1}(x)\right) + \frac{\pi}{2} - \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= \left(\frac{\pi}{2} + 2 \tan^{-1}(x)\right) + \left(\frac{\pi}{2} - 2 \tan^{-1}(x)\right)$$

$$= \pi$$

Hence, the value of $f\left(\frac{1}{2014}\right) = \pi$

183. We have

$$\begin{aligned} f(x) &= \sin^{-1}\left(\frac{6x}{x^2+9}\right) + 2 \tan^{-1}\left(-\frac{x}{3}\right) \\ &= \sin^{-1}\left(\frac{2 \cdot \left(\frac{x}{3}\right)}{1 + \left(\frac{x}{3}\right)^2}\right) - 2 \tan^{-1}\left(\frac{x}{3}\right) \\ &= 2 \tan^{-1}\left(\frac{x}{3}\right) - 2 \tan^{-1}\left(\frac{x}{3}\right) \\ &= 0 \end{aligned}$$

It will happen when $\left|\frac{x}{3}\right| \leq 1$

$$\Rightarrow |x| \leq 3$$

$$\Rightarrow -3 \leq x \leq 3$$

184. We have

$$\begin{aligned} f(x) &= \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + 2 \tan^{-1}(x) \\ &= 2 \tan^{-1}(x) + 2 \tan^{-1}(x), x \leq 0 \\ &= 0 \end{aligned}$$

It is possible only when $x \leq 0$

$$\Rightarrow x \in (-\infty, 0]$$

185. We have

$$\begin{aligned} f(x) &= 3 \cos^{-1}(2x^2 - 1) + 2 \cos^{-1}(4x^3 - 3x) \\ &= 3(2 \cos^{-1} x) + 2(2\pi - 3 \cos^{-1} x) \\ &\quad \left(\text{for } 0 \leq x < 1\right) \left(\text{for } -\frac{1}{2} \leq x \leq \frac{1}{2}\right) \end{aligned}$$

It is possible only when $0 \leq x \leq \frac{1}{2}$

$$\Rightarrow x \in \left[0, \frac{1}{2}\right]$$

Also,

$$\begin{aligned} f(x) &= 3 \cos^{-1}(2x^2 - 1) + 2 \cos^{-1}(4x^3 - 3x) \\ &= 3(2\pi - 2 \cos^{-1} x) + 2(-2\pi + 3 \cos^{-1} x) \\ &\quad \left(\text{for } 1 \leq x \leq 0\right) \left(\text{for } -1 \leq x < -\frac{1}{2}\right) \\ &= 2\pi \end{aligned}$$

It is possible only when $x \in \left[-1, -\frac{1}{2}\right]$

Hence, the value of x is

$$\left[-1, -\frac{1}{2}\right] \cup \left[0, \frac{1}{2}\right]$$

186. We have

$$\begin{aligned} \tan^{-1} y &= 4 \tan^{-1} x \\ \Rightarrow \tan^{-1} y &= \tan^{-1}\left(\frac{4x - 4x^3}{1 - 6x^2 + x^4}\right) \\ \Rightarrow y &= \frac{4x(1 - x^2)}{1 - 6x^2 + x^4} \end{aligned}$$

Which is a function of x .

$$\text{Let } \tan^{-1} x = \frac{\pi}{8}$$

$$\Rightarrow x = \tan\left(\frac{\pi}{8}\right)$$

$$\Rightarrow \tan^{-1} y = 4 \tan^{-1} x = \frac{\pi}{2}$$

$$\Rightarrow \frac{4x(1 - x^2)}{1 - 6x^2 + x^4} \rightarrow \infty$$

$$\Rightarrow 1 - 6x^2 + x^4 = 0$$

$$\Rightarrow x = \tan\left(\frac{\pi}{8}\right) \text{ is a root of } 1 + x^4 = 6x^2$$

187. Do yourself.

188. We have

$$\begin{aligned} \tan\left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} \frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{a}{b}\right) \\ = \tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right), \theta = \frac{1}{2} \cos^{-1} \left(\frac{a}{b}\right) \\ = \frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta}, \cos(2\theta) = \frac{a}{b} \\ = \frac{2(1 + \tan^2 \theta)}{1 - \tan^2 \theta} \\ = \frac{2}{\cos(2\theta)} \\ = \frac{2}{\frac{a}{b}} = \frac{2b}{a} \end{aligned}$$

189. Do yourself.

190. We have

$$\begin{aligned} \cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right) \\ = \tan^{-1}\left(\frac{a-b}{1+ab}\right) + \tan^{-1}\left(\frac{b-c}{1+bc}\right) + \tan^{-1}\left(\frac{c-a}{1+ca}\right) \\ = (\tan^{-1} a - \tan^{-1} b) + (\tan^{-1} b - \tan^{-1} c) + (\tan^{-1} c - \tan^{-1} a) \\ = 0 \end{aligned}$$

191. We have

$$\begin{aligned} \tan\left(\frac{1}{2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2} \cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right) \\ = \tan\left(\frac{1}{2} \cdot 2 \tan^{-1} x + \frac{1}{2} \cdot 2 \tan^{-1} y\right) \\ = \tan(\tan^{-1} x + \tan^{-1} y) \\ = \tan\left(\tan^{-1}\left(\frac{x+y}{1-xy}\right)\right) \\ = \left(\frac{x+y}{1-xy}\right), xy < 1 \end{aligned}$$

192. We have

$$\begin{aligned} \tan^{-1}\left(\frac{1-x}{1+x}\right) - \tan^{-1}\left(\frac{1-y}{1+y}\right) \\ = \{\tan^{-1}(1) - \tan^{-1}(x)\} - \{\tan^{-1}(1) - \tan^{-1}(y)\} \\ = \tan^{-1}(y) - \tan^{-1}(x) \end{aligned}$$

$$= \tan^{-1} \left(\frac{y-x}{1+xy} \right)$$

$$= \sin^{-1} \left(\frac{y-x}{\sqrt{1+x^2}\sqrt{1+y^2}} \right)$$

193. We have

$$\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A)$$

$$= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \tan^{-1} \left(\frac{\cot A + \cot^3 A}{1 - \cot^4 A} \right)$$

$$= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \tan^{-1} \left(\frac{\cot A}{1 - \cot^2 A} \right)$$

$$= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \tan^{-1} \left(\frac{\tan A}{\tan^2 A - 1} \right)$$

$$= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) + \tan^{-1} \left(-\frac{\tan A}{1 - \tan^2 A} \right)$$

$$= \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right) - \tan^{-1} \left(\frac{\tan A}{1 - \tan^2 A} \right)$$

$$= 0$$

194. We have

$$2 \tan^{-1} \left(\sqrt{\frac{a-b}{a+b}} \tan \left(\frac{\theta}{2} \right) \right)$$

$$= \cos^{-1} \left(\frac{1 - \left(\frac{a-b}{a+b} \right) \tan^2 \left(\frac{\theta}{2} \right)}{1 + \left(\frac{a-b}{a+b} \right) \tan^2 \left(\frac{\theta}{2} \right)} \right)$$

$$= \cos^{-1} \left(\frac{(a+b) - (a-b) \tan^2 \left(\frac{\theta}{2} \right)}{(a+b) + (a-b) \tan^2 \left(\frac{\theta}{2} \right)} \right)$$

$$= \cos^{-1} \left(\frac{a \left(1 - \tan^2 \left(\frac{\theta}{2} \right) \right) + b \left(1 + \tan^2 \left(\frac{\theta}{2} \right) \right)}{a \left(1 + \tan^2 \left(\frac{\theta}{2} \right) \right) + b \left(1 - \tan^2 \left(\frac{\theta}{2} \right) \right)} \right)$$

$$= \cos^{-1} \left(\frac{a \left(\frac{1 - \tan^2 \left(\frac{\theta}{2} \right)}{1 + \tan^2 \left(\frac{\theta}{2} \right)} \right) + b}{a + b \left(\frac{1 - \tan^2 \left(\frac{\theta}{2} \right)}{1 + \tan^2 \left(\frac{\theta}{2} \right)} \right)} \right)$$

$$= \cos^{-1} \left(\frac{a \cos \theta + b}{a + b \cos \theta} \right)$$

195. We have

$$2 \tan^{-1} (\tan^{-1} a + \tan^{-1} a^3)$$

$$= 2 \tan^{-1} \left(\tan^{-1} \left(\frac{a+a^3}{1-a^4} \right) \right)$$

$$= 2 \tan^{-1} \left(\tan^{-1} \left(\frac{a}{1-a^2} \right) \right)$$

$$= \left(\frac{2a}{1-a^2} \right)$$

$$= \tan (2 \tan^{-1} a)$$

196. Do yourself

197. Given, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$

Let $A = \sin^{-1} x, B = \sin^{-1} y, C = \sin^{-1} z$

$$\Rightarrow x = \sin A, y = \sin B, z = \sin C$$

$$\Rightarrow \cos A = \sqrt{1-x^2}, \cos B = \sqrt{1-y^2}$$

$$\text{and } \cos C = \sqrt{1-z^2}$$

$$\text{Now, } x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2}$$

$$= \sin A \cos A + \sin B \cos B + \sin C \cos C$$

$$= \frac{1}{2} [2 \sin A \cos A + 2 \sin B \cos B + 2 \sin C \cos C]$$

$$= \frac{1}{2} (\sin (2A) + \sin (2B) + \sin (2C))$$

$$= \frac{1}{2} (4 \sin A \sin B \sin C)$$

$$= (2 \sin A \sin B \sin C)$$

$$= 2xyz$$

198. We have

$$\cos^{-1} x \cos^{-1} y + \cos^{-1} z = \pi,$$

$$\Rightarrow \cos^{-1} x + \cos^{-1} y = \pi - \cos^{-1} z$$

$$\Rightarrow \cos^{-1} (x \cdot y \cdot \sqrt{1-x^2} \sqrt{1-y^2}) = \cos^{-1} (z)$$

$$\Rightarrow (xy + z)^2 = (\sqrt{(1-x^2)(1-y^2)})^2$$

$$\Rightarrow x^2 y^2 + 2xyz + z^2 = 1 - x^2 - y^2 + x^2 y^2$$

$$\Rightarrow 2xyz + z^2 = 1 - x^2 - y^2$$

$$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1$$

199. We have

$$\cos^{-1} \left(\frac{x}{2} \right) + \cos^{-1} \left(\frac{y}{3} \right) = \theta$$

$$\Rightarrow \cos^{-1} \left(\frac{xy}{6} - \sqrt{1-\frac{x^2}{4}} \sqrt{1-\frac{y^2}{9}} \right) = \theta$$

$$\Rightarrow \left(\frac{xy}{6} - \sqrt{1-\frac{x^2}{4}} \sqrt{1-\frac{y^2}{9}} \right) = \cos \theta$$

$$\Rightarrow \left(\frac{xy}{6} - \cos \theta \right)^2 = \left(1 - \frac{x^2}{4} \right) \left(1 - \frac{y^2}{9} \right)$$

$$\Rightarrow \frac{x^2 y^2}{36} - \frac{xy}{3} \cos \theta + \cos^2 \theta = 1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2 y^2}{36}$$

$$\Rightarrow -\frac{xy}{3} \cos \theta + \cos^2 \theta = 1 - \frac{x^2}{4} - \frac{y^2}{9}$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} - \frac{xy}{3} \cos \theta = 1 - \cos^2 \theta$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} - \frac{xy}{3} \cos \theta = \sin^2 \theta$$

$$\Rightarrow 9x^2 - 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta$$

Hence, the result.

200. Given, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

It is possible only when each term will provide us the maximum value

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2} = \sin^{-1} y = \sin^{-1} z$$

$$\text{So, } x = 1, y = 1, z = 1$$

Hence, the value of

$$\begin{aligned} x^2 + y^2 + z^2 - 2xyz \\ = 1 + 1 + 1 - 2 \\ = 1 \end{aligned}$$

201. Given, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

It is possible only when each term will provide us the maximum value

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2} = \sin^{-1} y = \sin^{-1} z$$

$$\text{So, } x = 1, y = 1, z = 1$$

Hence, the value of

$$\begin{aligned} xy + yz + zx &= 1 + 1 + 1 \\ &= 3 \end{aligned}$$

202. Given, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

It is possible only when each term will provide us the maximum value

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2} = \sin^{-1} y = \sin^{-1} z$$

$$\text{So, } x = 1, y = 1, z = 1$$

Hence, the value of

$$\begin{aligned} x^{2012} + y^{2012} + z^{2012} - \frac{9}{x^{2013} + y^{2013} + z^{2013}} \\ = 1 + 1 + 1 - \frac{9}{1 + 1 + 1} \\ = 3 - 3 \\ = 0 \end{aligned}$$

203. Given, $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$

It is possible only when, each term will provide us the maximum value.

$$\text{Thus, } \cos^{-1} x = \pi, \cos^{-1} y = \pi, \cos^{-1} z = \pi$$

$$x = -1, y = -1, z = -1$$

Hence, the value of

$$\begin{aligned} xy + yz + zx &= 1 + 1 + 1 \\ &= 3 \end{aligned}$$

204. Given, $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$

It is possible only when, each term will provide us the maximum value.

$$\text{Thus, } \cos^{-1} x = \pi, \cos^{-1} y = \pi, \cos^{-1} z = \pi$$

$$x = -1, y = -1, z = -1$$

Hence, the value of

$$\begin{aligned} \left(\frac{x^{2013} + y^{2013} + z^{2013} + 6}{x^{2014} + y^{2014} + z^{2014}} \right) \\ = \frac{1 - 1 - 1 + 6}{1 + 1 + 1} \\ = \frac{3}{3} = 1 \end{aligned}$$

205. Given, $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$

$$\Rightarrow \tan^{-1} x + \tan^{-1} y = \frac{\pi}{2} - \tan^{-1} z$$

$$\Rightarrow \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \cot^{-1}(z)$$

$$\Rightarrow \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \tan^{-1} \left(\frac{1}{z} \right)$$

$$\Rightarrow \left(\frac{x+y}{1-xy} \right) = \left(\frac{1}{z} \right)$$

$$\Rightarrow xz + yz = 1 - xy$$

$$\Rightarrow xy + yz + zx = 1$$

206. Given, $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \frac{\pi}{4}$$

$$\Rightarrow \left(\frac{x+y}{1-xy} \right) = \tan \left(\frac{\pi}{4} \right) = 1$$

$$\Rightarrow x + y = 1 - xy$$

$$\Rightarrow x + y + xy = 1$$

207. Given, $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$

$$\Rightarrow \tan^{-1} \left(\frac{x+y+z-xyz}{1-xy-yz-zx} \right) = \pi$$

$$\Rightarrow \left(\frac{x+y+z-xyz}{1-xy-yz-zx} \right) = \tan(\pi) = 0$$

$$\Rightarrow \left(\frac{x+y+z-xyz}{1-xy-yz-zx} \right) = 0$$

$$\Rightarrow x + y + z - xyz = 0$$

$$\Rightarrow x + y + z = xyz$$

Hence, the result.

208. Given, $\tan^{-1} \left(\frac{\sqrt{1+x^2} \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right) = \alpha$

$$\text{Put } x^2 = \sin(2\theta)$$

$$\text{Now, } \tan^{-1} \left(\frac{\sqrt{1+\sin 2\theta} \sqrt{1-\sin 2\theta}}{\sqrt{1+\sin 2\theta} + \sqrt{1-\sin 2\theta}} \right) = \alpha$$

$$\Rightarrow \tan^{-1} \left(\frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{(\cos \theta + \sin \theta) + (\cos \theta - \sin \theta)} \right) = \alpha$$

$$\Rightarrow \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right) = \alpha$$

$$\Rightarrow \tan^{-1}(\tan \theta) = \alpha$$

$$\Rightarrow \theta = \alpha$$

$$\Rightarrow 2\theta = 2\alpha$$

$$\Rightarrow \sin(2\theta) = \sin(2\alpha)$$

$$\Rightarrow x^2 = \sin(2\alpha)$$

209. We have

$$m = \tan^2(\sec^{-1} 2) + \cot^2(\operatorname{cosec}^{-1} 3)$$

$$= \tan^2(\theta) + \cot^2(\varphi), \text{ where}$$

$$\sec \theta = 2 \text{ and } \operatorname{cosec} \varphi = 3$$

$$= 1 + \sec^2 \theta + 1 + \operatorname{cosec}^2 \varphi$$

$$= 1 + 4 + 1 + 9$$

$$= 15$$

$$\text{Hence, the value of } (m^2 + m + 10)$$

$$= 225 + 15 + 10$$

$$= 250$$

$$210. \text{ Given, } \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = \frac{\pi}{4}$$

$$\Rightarrow \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = \frac{\pi}{2}$$

$$\Rightarrow \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = 1$$

$$\Rightarrow 3 \sin 2\theta = 5 + 4 \cos 2\theta$$

$$\Rightarrow \frac{3 \cdot 2 \tan \theta}{1 + \tan^2 \theta} = 5 + 4 \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)$$

$$\Rightarrow \frac{6 \tan \theta}{1 + \tan^2 \theta} = \frac{5 + 5 \tan^2 \theta + 4 - 4 \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\Rightarrow 6 \tan \theta = \tan^2 \theta + 9$$

$$\Rightarrow \tan^2 \theta - 6 \tan \theta + 9 = 0$$

$$\Rightarrow (\tan \theta - 3)^2 = 0$$

$$\Rightarrow (\tan \theta - 3) = 0$$

$$\Rightarrow \tan \theta = 3$$

$$211. \text{ Given, } m = \frac{(\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3)}{(\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3)}$$

$$= \frac{\pi}{\frac{\pi}{2}} = 2$$

Hence, the value of

$$(m + 2)^{m+1} = (2 + 2)^3 = 4^3 = 64$$

212. Given equation is

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{3\pi}{4}$$

$$\Rightarrow \tan^{-1} \left(\frac{2x + 3x}{1 - 6x^2} \right) = \frac{\pi}{4}$$

$$\Rightarrow \left(\frac{5x}{1 - 6x^2} \right) = 1$$

$$\Rightarrow 5x = 1 - 6x^2$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow 6x^2 + 6x - x - 1 = 0$$

$$\Rightarrow 6x(x + 1) - 1(x + 1) = 0$$

$$\Rightarrow (6x - 1)(x + 1) = 0$$

$$\Rightarrow x = 1, \frac{1}{6}$$

$$213. \tan^{-1} \left(\frac{x+1}{x-1} \right) + \tan^{-1} \left(\frac{x-1}{x} \right) = \tan^{-1}(7)$$

$$\Rightarrow \tan^{-1} \left(\frac{\frac{x+1}{x-1} + \frac{x-1}{x}}{1 - \frac{x+1}{x-1} \times \frac{x-1}{x}} \right) = \tan^{-1}(7)$$

$$\Rightarrow \tan^{-1} \left(\frac{x^2 + x + x^2 - 2x + 1}{x^2 - x - x^2 + 1} \right) = \tan^{-1}(7)$$

$$\Rightarrow \tan^{-1} \left(\frac{2x^2 - x + 1}{1 - x} \right) = \tan^{-1}(7)$$

$$\Rightarrow \left(\frac{2x^2 - x + 1}{1 - x} \right) = 7$$

$$\Rightarrow 2x^2 - x + 1 = -7 + 7x$$

$$\Rightarrow 2x^2 - 8x + 8 = 0$$

$$\Rightarrow x^2 - 4x + 4 = 0$$

$$\Rightarrow (x - 2)^2 = 0$$

$$\Rightarrow x = 2$$

Hence, the solution is $x = 2$

214. Given equation is

$$\sin^{-1}(2x) + \sin^{-1}(x) = \frac{\pi}{3}$$

$$\Rightarrow \sin^{-1}(2x) = \frac{\pi}{3} - \sin^{-1}x$$

$$\Rightarrow \sin^{-1}(2x) = \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1}x$$

$$\Rightarrow \sin^{-1}(2x) = \sin^{-1} \left(\frac{\sqrt{3}}{2} \sqrt{1-x^2} - x \sqrt{1 - \frac{3}{4}} \right)$$

$$\Rightarrow (2x) = \left(\frac{\sqrt{3}}{2} \sqrt{1-x^2} - x \sqrt{1 - \frac{3}{4}} \right)$$

$$\Rightarrow (2x) = \left(\frac{\sqrt{3}}{2} \sqrt{1-x^2} - \frac{x}{2} \right)$$

$$\Rightarrow \left(2x + \frac{x}{2} \right) = \frac{\sqrt{3}}{2} \sqrt{1-x^2}$$

$$\Rightarrow \left(\frac{5x}{2} \right) = \frac{\sqrt{3}}{2} \sqrt{1-x^2}$$

$$\Rightarrow \left(\frac{25x^2}{4} \right) = \frac{3(1-x^2)}{4}$$

$$\Rightarrow 25x^2 = 3 - 3x^2$$

$$\Rightarrow 28x^2 = 3$$

$$\Rightarrow x^2 = \frac{3}{28}$$

$$\Rightarrow x = \pm \sqrt{\frac{3}{28}} = \pm \frac{\sqrt{3}}{2\sqrt{7}}$$

215. Given equation is

$$\begin{aligned} \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}x &= \frac{\pi}{4} \\ \Rightarrow \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) + \cos^{-1}x &= \frac{\pi}{4} \\ \Rightarrow \cos^{-1}x &= \frac{\pi}{4} - \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) \\ \Rightarrow \cos^{-1}x &= \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) - \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) \\ \Rightarrow \cos^{-1}x &= \cos^{-1}\left(\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{5}} + \sqrt{1 - \frac{1}{2}} \sqrt{1 - \frac{4}{5}}\right) \\ \Rightarrow x &= \left(\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{5}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{5}}\right) \\ \Rightarrow x &= \frac{3}{\sqrt{10}} \end{aligned}$$

216. Given equation is

$$\begin{aligned} \sin^{-1}(x) + \sin^{-1}(3x) &= \frac{\pi}{3} \\ \Rightarrow \sin^{-1}(3x) &= \frac{\pi}{3} - \sin^{-1}(x) \\ \Rightarrow \sin^{-1}(3x) &= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}(x) \\ \Rightarrow \sin^{-1}(3x) &= \sin^{-1}\left(\frac{\sqrt{3}}{2} \sqrt{1-x^2} - x \sqrt{1-\frac{3}{4}}\right) \\ \Rightarrow (3x) &= \left(\frac{\sqrt{3}}{2} \sqrt{1-x^2} - x \sqrt{1-\frac{3}{4}}\right) \\ \Rightarrow \left(3x + \frac{x}{2}\right) &= \frac{\sqrt{3}}{2} \sqrt{1-x^2} \\ \Rightarrow \frac{7x}{2} &= \frac{\sqrt{3}}{2} \sqrt{1-x^2} \\ \Rightarrow 49x^2 &= 3(1-x^2) \\ \Rightarrow 52x^2 &= 3 \\ \Rightarrow x^2 &= \frac{3}{52} \\ \Rightarrow x &= \pm \sqrt{\frac{3}{52}} \end{aligned}$$

Hence, the solutions are

$$\left\{-\sqrt{\frac{3}{52}}, \sqrt{\frac{3}{52}}\right\}$$

217. Given equation is

$$\begin{aligned} \tan^{-1}\left(\frac{1}{1+2x}\right) + \tan^{-1}\left(\frac{1}{1+4x}\right) &= \tan^{-1}\left(\frac{2}{x^2}\right) \\ \Rightarrow \tan^{-1}\left(\frac{\frac{1}{1+2x} + \frac{1}{1+4x}}{1 - \frac{1}{1+2x} \times \frac{1}{1+4x}}\right) &= \tan^{-1}\left(\frac{2}{x^2}\right) \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{\frac{1}{1+2x} + \frac{1}{1+4x}}{1 - \frac{1}{1+2x} \times \frac{1}{1+4x}}\right) &= \left(\frac{2}{x^2}\right) \\ \Rightarrow \left(\frac{1+2x+1+4x}{1+6x+8x^2-1}\right) &= \left(\frac{2}{x^2}\right) \\ \Rightarrow \left(\frac{2+6x}{6x+8x^2}\right) &= \left(\frac{2}{x^2}\right) \\ \Rightarrow \frac{1+3x}{3x+4x^2} &= \frac{2}{x^2} \\ \Rightarrow 3x^3+x^2 &= 6x+8x^2 \\ \Rightarrow 3x^3-7x^2-6x &= 0 \\ \Rightarrow (3x^2-7x-6)x &= 0 \\ \Rightarrow (3x^2-9x+2x-6)x &= 0 \\ \Rightarrow x(x-3)(2x+3) &= 0 \\ \Rightarrow x &= 0, 3, -\frac{3}{2} \end{aligned}$$

Hence, the solutions are

$$\left\{0, 3, -\frac{3}{2}\right\}$$

218. Given equation is

$$\begin{aligned} 2 \tan^{-1}(2x+1) &= \cos^{-1}x \\ \Rightarrow \tan^{-1}\left(\frac{2(2x+1)}{1-(2x+1)^2}\right) &= \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) \\ \Rightarrow \frac{2(2x+1)}{1-(2x+1)^2} &= \left(\frac{\sqrt{1-x^2}}{x}\right) \\ \Rightarrow \frac{2(2x+1)}{(4x^2+4x)} &= \left(\frac{\sqrt{1-x^2}}{x}\right) \\ \Rightarrow \frac{(2x+1)}{(2x^2+2x)} &= \left(\frac{\sqrt{1-x^2}}{x}\right), x=0 \\ \Rightarrow \frac{(2x+1)}{(2x+2)} &= \sqrt{1-x^2}, x=0 \\ \Rightarrow \frac{(2x+1)^2}{(2x+2)^2} &= (1-x^2), x=0 \\ \Rightarrow \frac{4x^2+4x+1}{4x^2+8x+4} &= 1-x^2, x=0 \\ \Rightarrow 3-4x^4-8x^3-4x^2+4x &= 0, x=0 \\ \Rightarrow 4x^4+8x^3+4x^2-4x-3 &= 0, x=0 \end{aligned}$$

Clearly, it has 3 solutions.

219. Given equation is

$$\begin{aligned} \cos^{-1}x - \sin^{-1}x &= \cos^{-1}(x\sqrt{3}) \\ \Rightarrow \cos^{-1}x - \frac{\pi}{2} + \cos^{-1}x &= \cos^{-1}(x\sqrt{3}) \\ \Rightarrow 2 \cos^{-1}x - \frac{\pi}{2} &= \cos^{-1}(x\sqrt{3}) \end{aligned}$$